



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**Faculty of
Computing**

**SEMESTER II 2025/2026 SECD2613-04
SYSTEM ANALYSIS AND DESIGN**

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GROUP PROJECT PHASE 3 - GROUP 2

GROUP PROJECT : UTM TRANSPORT SYSTEM

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UTM Transport App :

<https://www.figma.com/proto/izkx4m33e5fG88ATDpNall/Untitled?node-id=4-8&viewport=3274%2C-492%2C0.58&t=jQBN87iotmX2mFmr-1&scaling=scale-down&content-scaling=fixed&starting-point-node-id=4%3A8&page-id=0%3A1>

Admin Portal:

<https://www.figma.com/proto/8yDROj3rdX8zbeU1EIM0mA/ADMIN-PORTAL-UTM-TRANSPORT?node-id=26-2&p=f&t=WYRWbv3dzfGRXAJ5-1&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=26%3A2>

Prototype Demo Link:

<https://youtu.be/pYzF1KddwCs>

Github Link:

https://github.com/Dc-by1/Group2_Project1_SAD_20252026/tree/main

TABLE OF CONTENT

1.0 Overview of the Project.....	1
2.0 Problem Statement.....	2
3.0 Proposed Solutions.....	4
4.0 Current Business Process / Workflow.....	6
5.0 Logical DFD AS-IS System.....	7
6.0 System Analysis and Specification.....	10
6.1 Logical DFD TO-BE System.....	10
6.2 Process Specification (based on Logical DFD TO-BE).....	13
7.0 Physical System Design.....	27
7.1 Physical DFD TO-BE System.....	27
Diagram 0.....	27
Child Diagram.....	28
Partitioning.....	29
CRUD Matrix.....	30
Event Response Table.....	31
Structure Chart.....	33
System Architecture.....	34
8.0 System Wireframe.....	38
9.0 Summary of the Proposed System.....	61

1.0 Overview of the Project

Universiti Teknologi Malaysia, as one of the best technological universities in the world, has many students who move around every day from their residences to the various faculties. Therefore, efficient campus transportation is no longer just a convenience but a vital necessity. The university operates an internal shuttle bus service running different routes within the university campus connecting various residential colleges, faculties, and administration blocks.

However, the current bus service is not satisfactory to the students due to several reasons such as the lack of active buses with high student density. These issues lead to the inconvenience when using the bus service provided and may affect the daily study plan such as students being unable to attend the class on time due to the delay of the buses.

Thus, this project is to design and develop a new UTM Transport App to ensure that students can get transportation easily on campus. This mobile application focuses on bringing the physical and virtual sides of campus life together through the use of real-time data analysis and mobile applications to maximize efficiency and engagement within the campus ecosystem.

In this project, we will collect the students' suggestions or even complaints, and analyse the pain points of the existing transport system. After that, the existing features in the current system will be modified and the new features will be created based on students' requirements. All of these features will be included in our UTM Transport App. Therefore, instead of switching various mobile applications, students can get satisfactory transportation by using only a single mobile application. We will also develop an admin portal for UTM Administrator to support our application to ensure we have a complete UTM Transport System.

2.0 Problem Statement

At a campus as huge as Universiti Teknologi Malaysia (UTM), transportation is not just a "nice-to-have" feature, it is the heartbeat of our daily academic lives. However, the current transportation system tends to be fragmented and traditional that just cannot keep up with the intense schedule of a modern student. For most of us, arranging our traveling schedule is not the smooth transition it should be. It has become a daily source of stress and friction that drains our focus and productivity. To alleviate this issue, we need to address five specific "pain points" that are currently making life harder for the UTM community:

1. The Bus Arrival Uncertainty

Right now, the biggest headache is the total lack of transparency. We are forced to stand at bus stops for who-knows-how-long, often sweating under the scorching sun or getting soaked in a heavy rain, without any idea if the bus is two minutes away or twenty. This uncertainty causes a domino effect, such that we end up late for crucial lectures, miss the start of lab sessions, and live in a constant state of "transit anxiety" that makes it impossible to plan our study groups or even a simple lunch break effectively.

2. The Peak-Hour Seat Shortages

During the morning and evening rush, the campus transport system basically turns into "survival of the fittest" mode. This is because we have no way to check how full a bus is in advance, we often have to watch bus after bus drive past because they are packed with people from the previous stations. It is not just about the delay, but also about the discomfort of being squeezed into overcrowded vehicles and the safety risks that come with it, leaving those at later stops completely stranded.

3. The Inefficient Carpooling

So many of us are willing to share rides or lend a helping hand to ease the load on the buses, but there is not a safe, official way to make that happen. We are currently stuck using disorganized social media chats or word-of-mouth, which is awkward and unreliable. This "fragmented carpooling" means thousands of seats go empty every day while other students are left waiting at the station or walking with their bare feet, wasting a huge opportunity to build a better campus community and lower our carbon footprint.

4. No Route Planning Tools

UTM is huge, and for "freshies" or visitors, it can feel like a total maze. Without a smart tool to help plan a route, students struggle to find the fastest way to get from their residential colleges to distant faculties or admin buildings. We end up wasting time taking long, inefficient paths or walking much further than we need to, simply because we do not have a map that shows us how to navigate the transit transfers correctly.

5. The Lack of Feedback

The most frustrating part of all this is that when things go wrong, we have no real voice. There is not a dedicated, systematic way to report things like a bus being consistently late, a vehicle being dirty, or safety concerns with driving. Without this feedback, the university admin is essentially "flying blind," making it impossible for them to see where the service is failing or how to fix it. This silence leads to a cycle of the same problems happening semester after semester with no end in sight.

3.0 Proposed Solutions

- **Design a Secure User Management Module**

This module focuses on account management functions such as account creation, authentication, and user-profile maintenance to ensure that users of the application are indeed verified UTM students. They need to sign up using UTM email. So, users just need to register their account and click on the "Remember Me" checkbox if one wants quick future logins.

- **Implement a Live Tracking & Mapping Module**

This module offers users the ability to view live information on campus buses' routes and locations. The information displayed consists of the actual location of the bus on the map, along with an Estimated Time of Arrival (ETA) at their current station. The module enables users to pick any bus on the map and observe its movement, along with its plate number.

- **Implement a Seat Availability Monitor Module**

This module allows users to view real-time seat availability of approaching campus buses. The information displayed includes the current occupancy level (available, moderate, or full). So, users can select a bus and check its seat status along with its route, helping them decide whether to board or wait for the next bus.

- **Create a Carpool Module**

This module helps in making travelling on the campus flexible since it offers the students the ability to form rides with other students. Users have two options in this module either to form their own carpool ride or join others. Users can go through the list of available rides and find someone who matches.

- **Design a Route Planning Module**

This module acts as the smart guide in planning how users can travel on the campus in the best way. The fastest route options will be provided together with step by step instructions as well as any required bus transfer along the way. Users can simply input the destination point from the point of departure to access various route options.

- **Implement a Feedback & Rating Module**

This module allows the users to help in improving campus transport through feedback. Here, users can rate their experience riding the bus and report problems such as lateness and overcrowding among others. Once the trip ends, a rating window pops out where users can rate the bus.

4.0 Current Business Process / Workflow

Currently, navigating the huge Universiti Teknologi Malaysia (UTM) campus relies heavily on traditional, manual coordination and fixed timetable shuttle buses. When a student needs to travel from their residential college to a faculty building, they walk to the nearest designated bus stop and wait. Due to there is no live tracking visibility into the transit network, the student must rely on guesswork, often standing under the scorching hot sun or getting soaked during sudden heavy rains, unsure if the bus is two minutes or twenty minutes away.

During peak morning and evening hours, the situation deteriorates. When a bus finally arrives, the student frequently discovers it is already packed with passengers from the previous stops. Unable to board, the student is forced to wait for the next uncertain bus, resulting in a domino effect of missed lectures, late lab sessions, and high “transit anxiety”. If a student decides to seek alternative transport, they must resort to fragmented, unofficial carpooling. They open disorganized social media groups or messaging apps (like Telegram) to manually ask for rides, which is inefficient and poses security concerns as driver identities are not formally verified.

Furthermore, for freshmen or visitors navigating the campus maze, there is no centralized tool to plan their journey. They often take long, inefficient walking paths or miss necessary bus transfers. Finally, when the journey concludes, if the students experience a delayed schedule, reckless driving, or a dirty vehicle, there is no dedicated channel to report it. The administration remains unaware of these daily operational failures, and the cycle of frustration continues into the next semester.

5.0 Logical DFD AS-IS System

Context Diagram

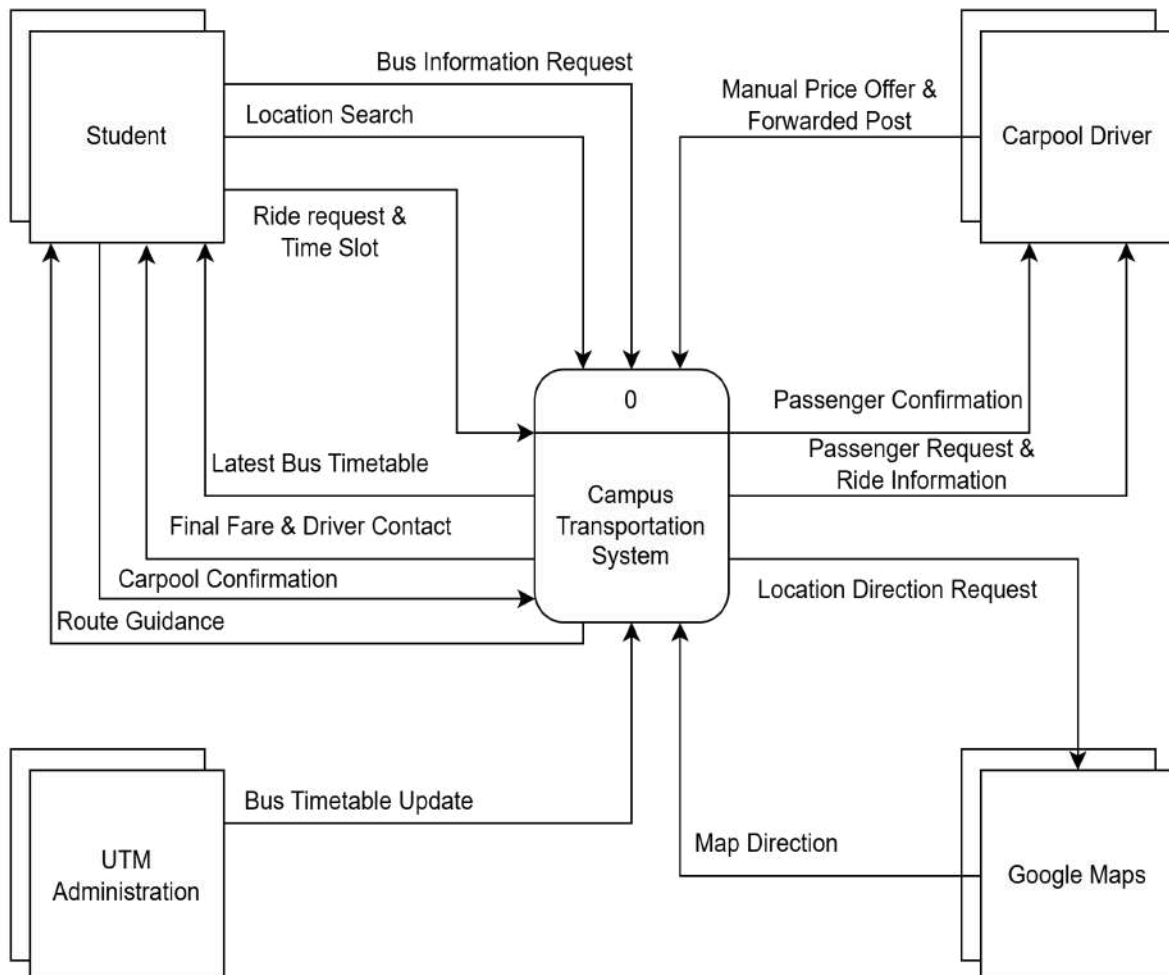
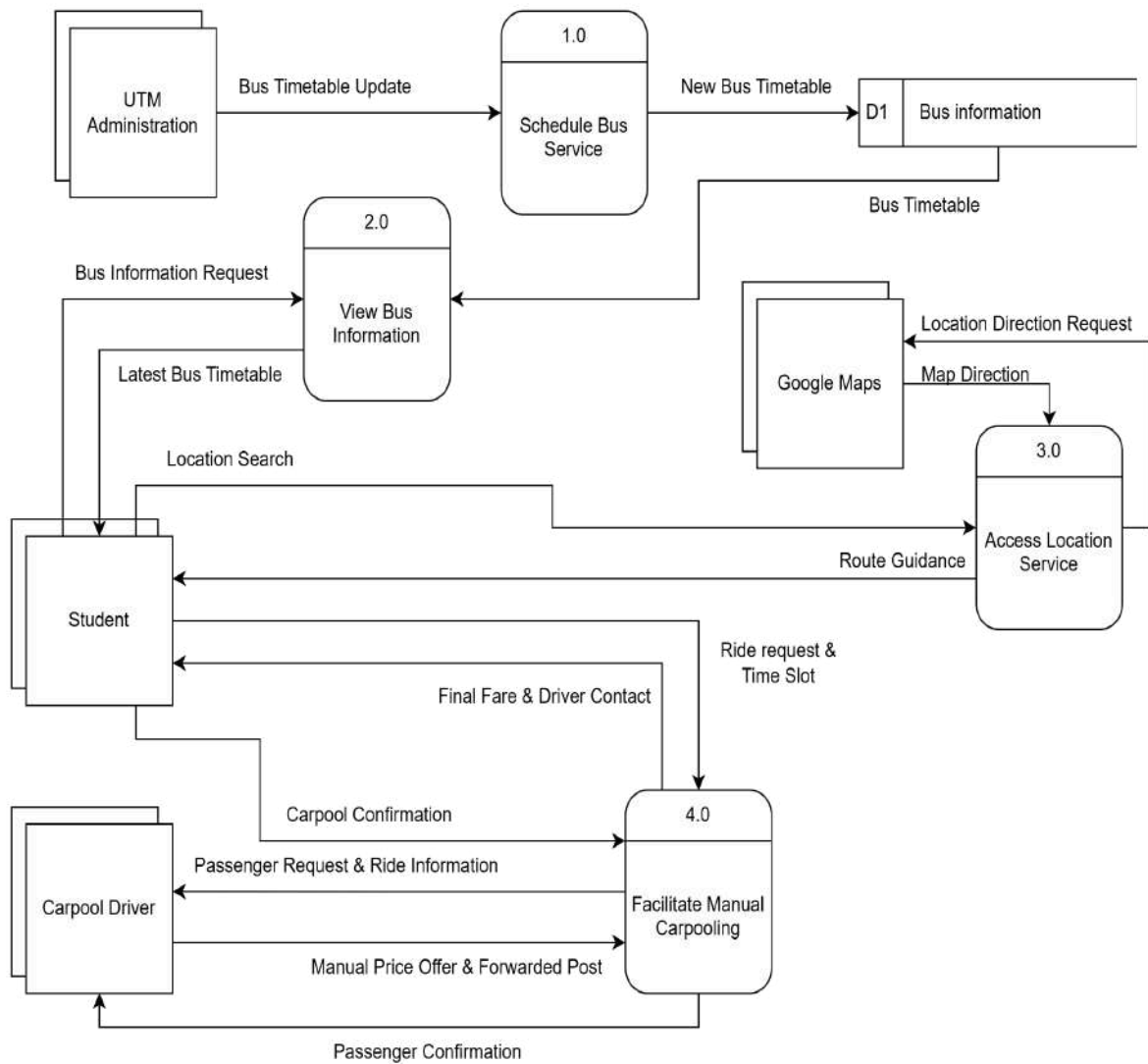
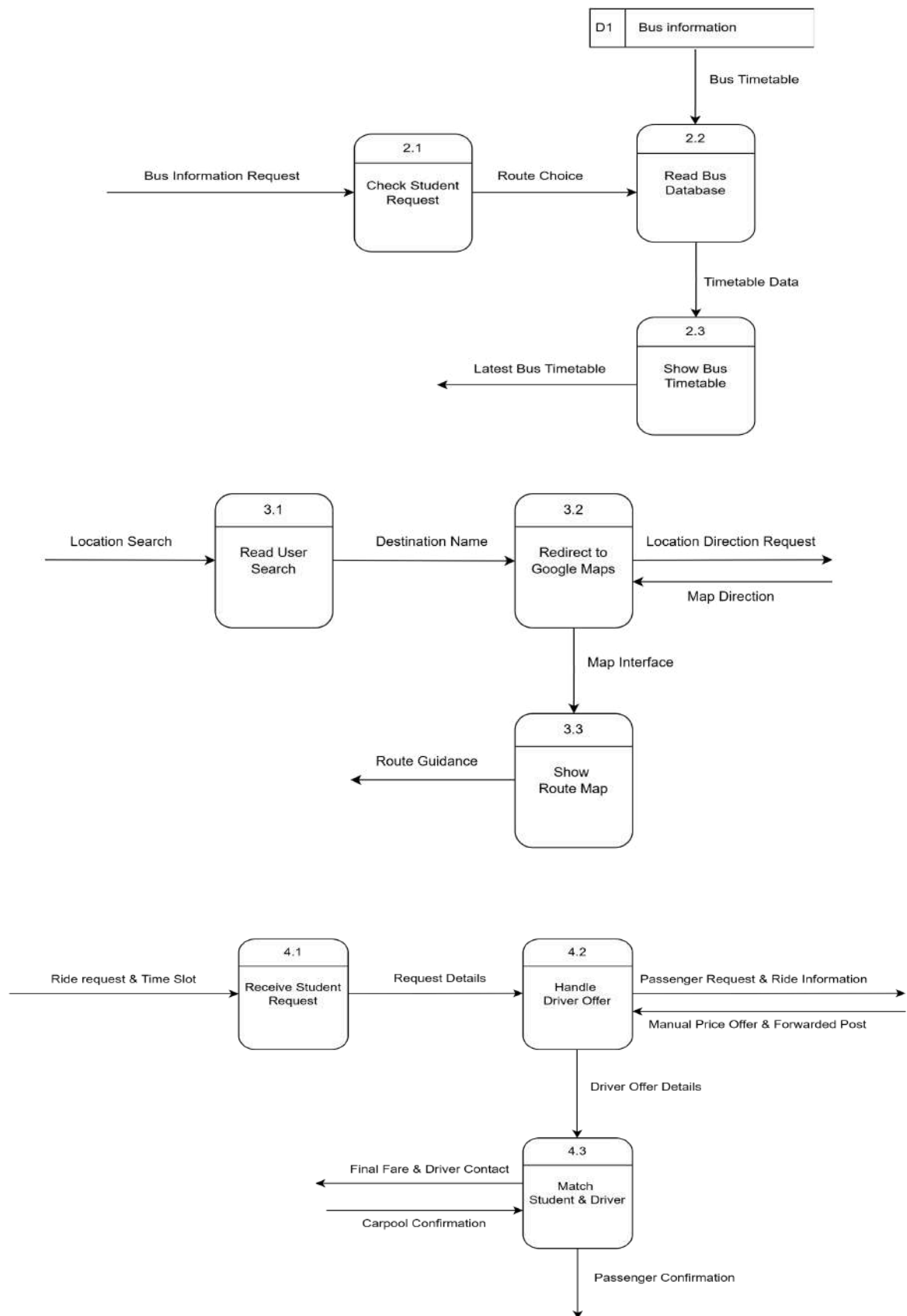


Diagram 0



Child Diagram



6.0 System Analysis and Specification

6.1 Logical DFD TO-BE System

Context Diagram

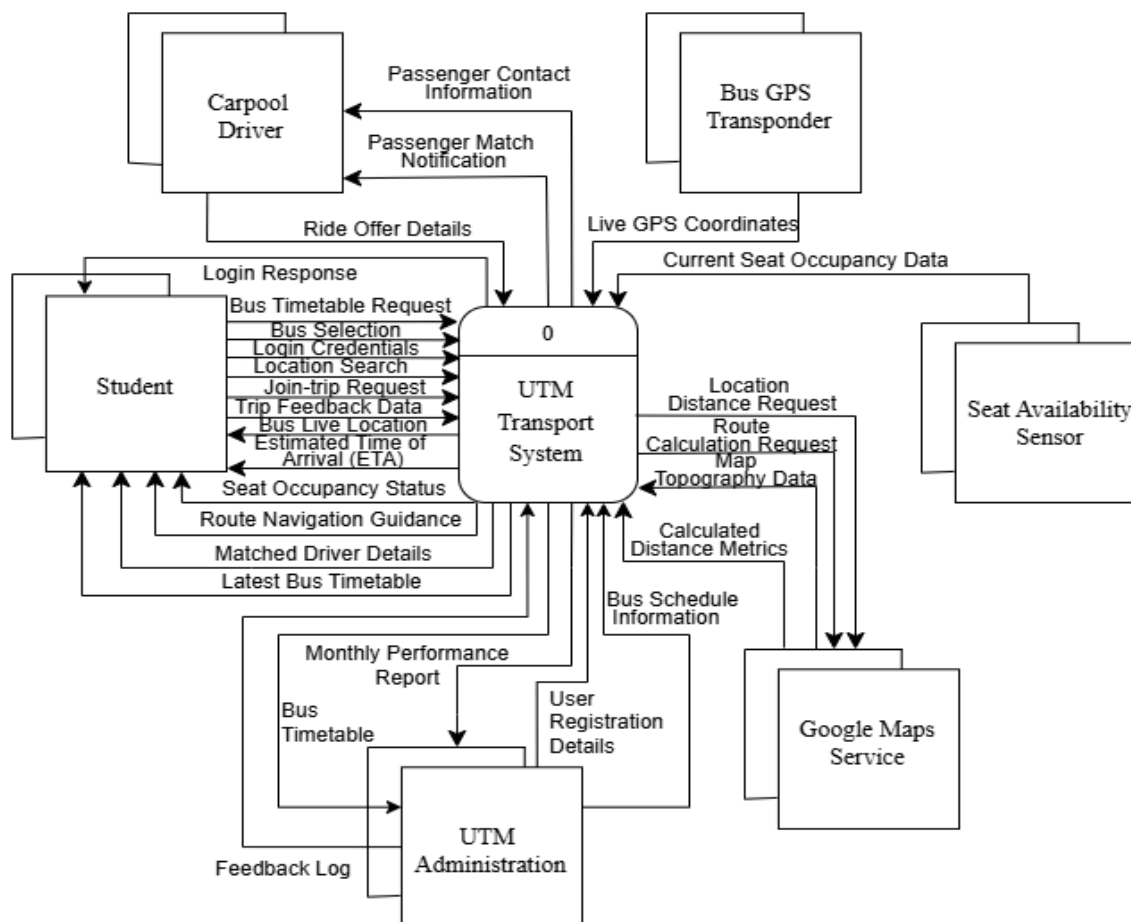
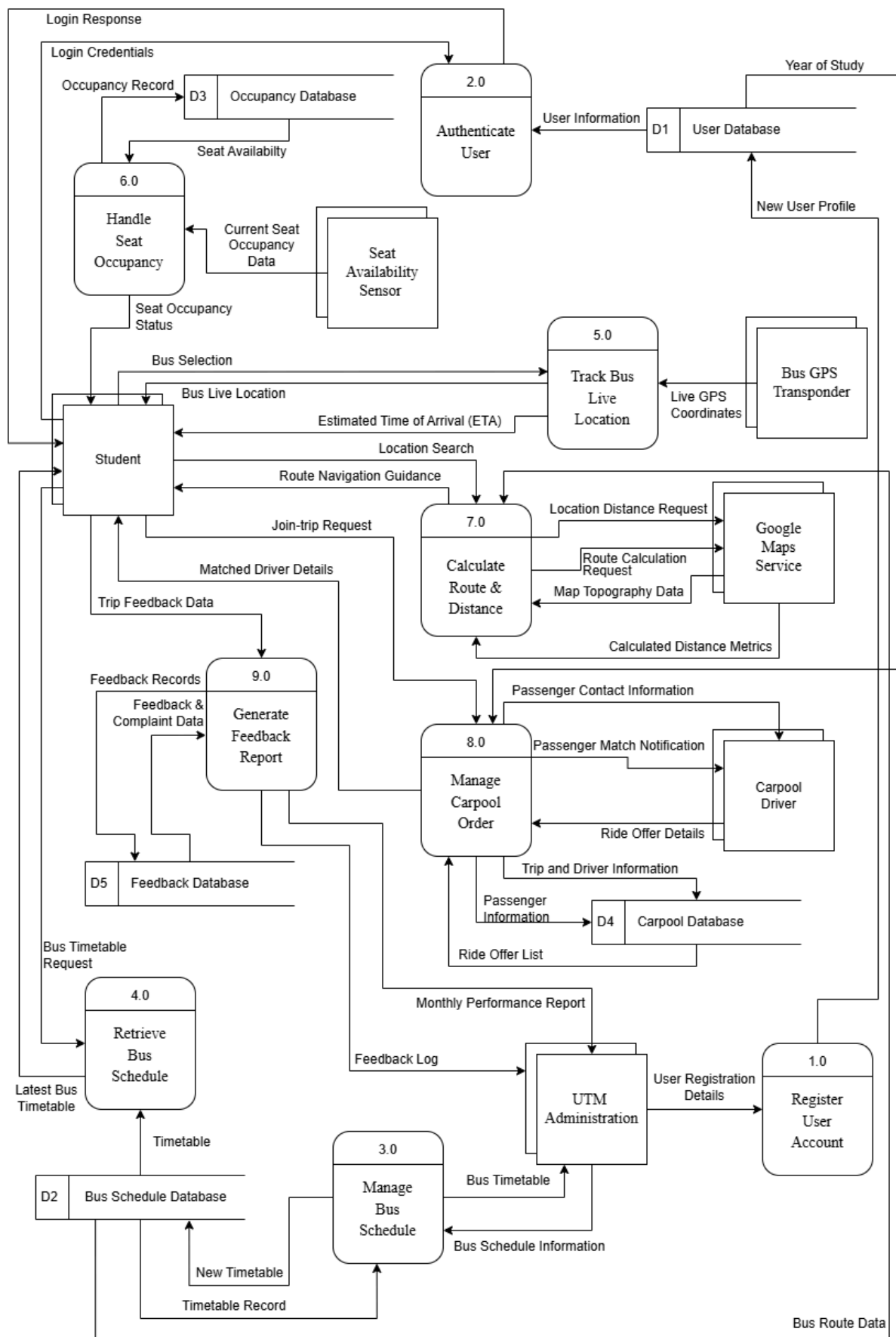
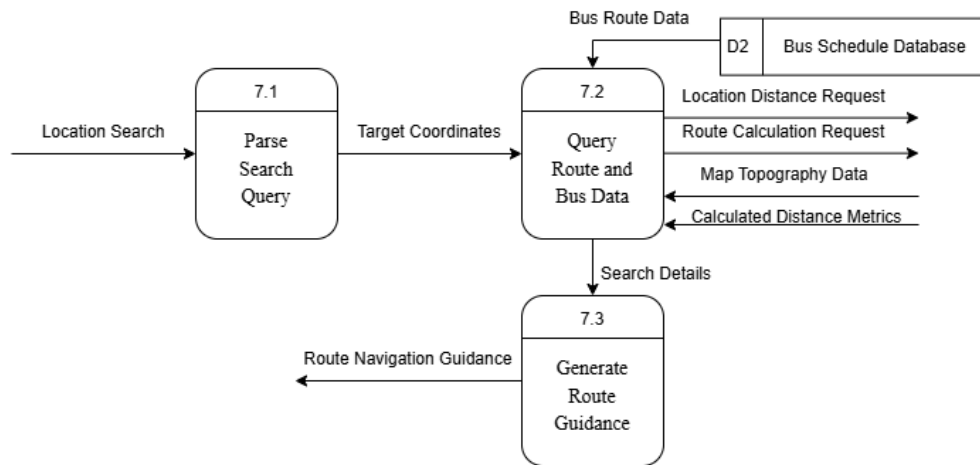


Diagram 0

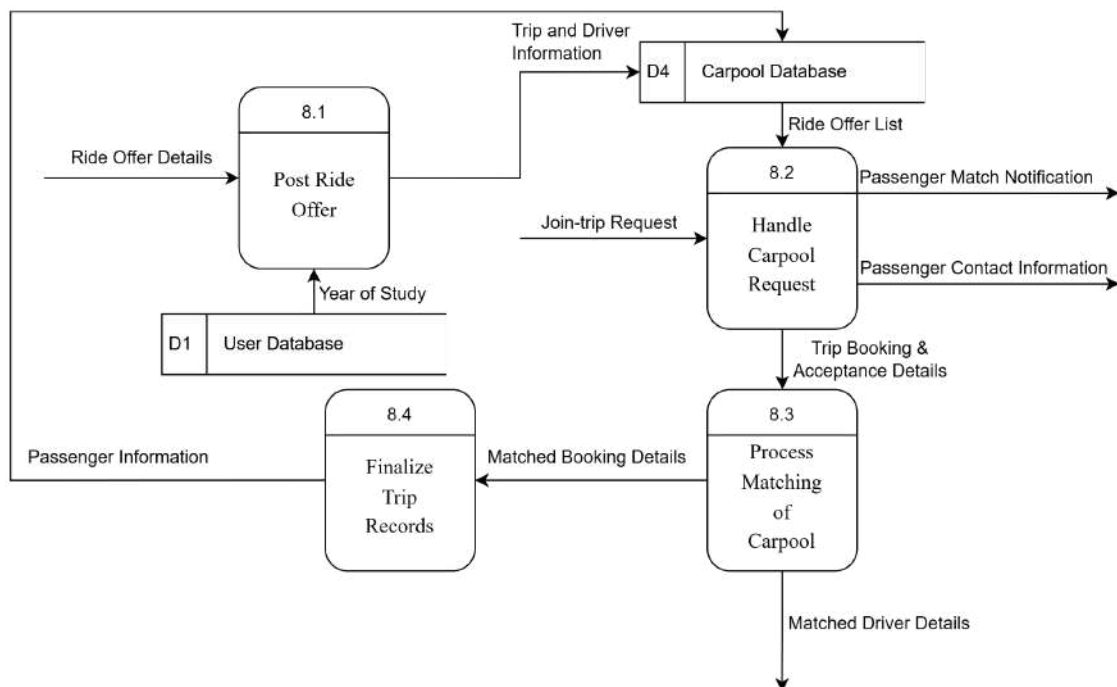


Child Diagram

Process 7.1, 7.2 and 7.3: Calculate Route & Distance



Process 8.1, 8.2, 8.3 and 8.4: Carpool



6.2 Process Specification (based on Logical DFD TO-BE)

Number : 1.0 Name : Register User Account Description : UTM Administration inputs the student's registration details to create a new user profile in the User Database.	
Input Data Flow : User Registration Details from UTM Administration	
Output Data Flow : New User Profile to User Database	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ User Registration Details from UTM Administration IF Student ID is empty OR Temporary Login Credential is empty OR Email is empty THEN RETURN Registration Error to UTM Administration ELSE CHECK Student ID in User Database IF Student ID exists THEN RETURN Registration Error to UTM Administration ELSE CREATE New User Profile WRITE New User Profile to User Database RETURN Registration Successful to UTM Administration ENDIF ENDIF	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 2.0 Name : Authenticate User Description : Receives the student's login credentials, then checks with user information in User Database and lastly returns a login access to the student.	
Input Data Flow : Login Credentials from Student User Information from User Database	
Output Data Flow : Login Access to Student	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ Login Credentials from Student FIND User Information in User Database IF Login Credentials matched with User Information THEN RETURN Successful Login Access to Student ELSE RETURN Failed Login Access to Student ENDIF	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 3.0 Name : Manage Bus Schedule Description : Receives bus timetable information from UTM Administration, checks with an existing timetable record in the Bus Schedule Database, and creates a new record or updates the existing record accordingly.	
Input Data Flow : Bus Timetable Information from UTM Administration Timetable Record from Bus Schedule Database	
Output Data Flow : New Timetable to Bus Schedule Database Bus Timetable to UTM Administration	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ Bus Timetable Information from UTM Administration RETRIEVE Timetable Record from Bus Schedule Database SEND Bus Timetable to UTM Administration IF Bus Timetable is found IF UTM Administration request to delete bus schedule THEN DELETE Bus Timetable from Bus Schedule Database DISPLAY "Bus Schedule Deleted Successfully" ELSE IF UTM Administration request to update bus schedule THEN UPDATE Bus Timetable with new information WRITE New Timetable to Bus Schedule Database ELSE IF UTM Administration request to create bus schedule THEN CREATE New Timetable Record WRITE New Timetable to Bus Schedule Database ELSE continue ENDIF ELSE SEND error message to UTM Administration ENDIF	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 4.0 Name : Retrieve Bus Schedule Description : Retrieves bus timetable information from the Bus Schedule Database and provides it to the student after student request.	
Input Data Flow : Bus Timetable Request from Student Timetable from Bus Schedule Database	
Output Data Flow : Latest Bus Timetable to Student	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ Bus Timetable Request from Student RETRIEVE Timetable from Bus Schedule Database IF Timetable is found THEN SEND Latest Bus Timetable to Student ELSE SEND error message to Student ENDIF	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 5.0 Name : Track Bus Live Location Description : Retrieve real-time GPS coordinates from the bus transponder and provide live location and ETA updates to the student.	
Input Data Flow : Bus Selection from Student Live GPS Coordinates from GPS Transponder	
Output Data Flow : Bus Live Location to Student Estimated Time of Arrival to Student	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: GET Bus Selection from Student DO WHILE Bus is Active GET Live GPS Coordinates from GPS Transponder IF GPS Coordinates are available THEN CALCULATE Estimated Time of Arrival SEND Bus Live Location to Student SEND Estimated Time of Arrival to Student ELSE SEND "GPS signal unavailable" to Student ENDIF ENDDO	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 6.0

Name : Handle Seat Occupancy

Description : Receive real-time seat occupancy data from the seat availability sensor, update occupancy record in the database, and return the seat occupancy status to Student.

Input Data Flow :

Current Seat Occupancy Data from Seat Availability Sensor

Seat Availability from Occupancy Database

Output Data Flow :

Occupancy Record to Occupancy Database

Seat Occupancy Status to Student

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

DO WHILE system is operating

 GET Current Seat Occupancy Data from Seat Availability Sensor

 IF Seat Availability Sensor detects passenger boarding

 THEN INCREASE occupancy count by 1

 ELSE IF Sensor detects passenger alighting

 THEN DECREASE occupancy count by 1

 ELSE

 REMAIN current occupancy count

 ENDIF

 MOVE occupancy count to Occupancy Record

 WRITE Occupancy Record to Occupancy Database

 READ Seat Availability from Occupancy Database

 IF Seat Availability ≥ 15

 THEN SET Seat Occupancy Status to Available

 ELSE IF Seat Availability > 1 AND Seat Availability < 15

 THEN SET Seat Occupancy Status to Moderate

 ELSE

 SET Seat Occupancy Status to Full

 ENDIF

 SEND Seat Occupancy Status to Student

ENDDO

Refer to: Name: _____

☒ Structured English

☐ Decision Table

☐ Decision Tree

Unresolved Issues: -

Number : 7.1 Name : Parse Search Query Description : Parse the student's location search input into structured target coordinates	
Input Data Flow : Location Search from Student	
Output Data Flow : Target Coordinates to Process 7.2	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: GET Location Search from Student IF Location Search is NOT empty THEN PARSE location name into Target Coordinates (latitude, longitude) SEND Target Coordinates to Process 7.2 ELSE SEND error message to Student ENDIF	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 7.2

Name : Query Route and Bus Data

Description : Queries the Google Maps Service using the target coordinates to retrieve route and distance data, and retrieves available campus bus routes from the Bus Schedule Database that serve the requested destination.

Input Data Flow :

Target Coordinates from Process 7.1

Map Topography Data from Google Maps Service

Calculated Distance Metrics from Google Maps Service

Bus Route Data from Bus Schedule Database

Output Data Flow :

Location Distance Request to Google Maps Service

Route Calculation Request to Google Maps Service

Search Details to Process 7.3

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

GET Target Coordinates from Process 7.1

RETRIEVE Bus Route Data from Bus Schedule Database

IF Bus Route Data covers Target Coordinates

 THEN SEND Location Distance Request to Google Maps Service based on available Bus Route Data

 SEND Route Calculation Request to Google Maps Service based on available Bus Route Data

 RECEIVE Map Topography Data from Google Maps Service

 RECEIVE Calculated Distance Metrics from Google Maps Service

 SET Search Details with Map Topography Data, Calculated Distance Metrics and Bus Route Data

 SEND Search Details to Process 7.3

ELSE

 SEND error message to Student

ENDIF

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 7.3 Name : Generate Route Guidance Description : Processes the combined route and bus data to generate complete route navigation guidance, including directions, distances, and available campus bus options, and delivers it to the student.	
Input Data Flow : Search Details from Process 7.2	
Output Data Flow : Route Navigation Guidance to Student	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: GET Search Details from Process 7.2 SET Route Navigation Guidance with directions, distance and available bus information SEND Route Navigation Guidance to Student	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 8.1

Name : Post Ride Offer

Description : Verifies student's year of study, then receives ride details from a student driver that offers a carpool, then validates the details, and stores the offer in the carpool database.

Input Data Flow :

Ride Offer Details from Carpool Driver

Year of Study from User Database

Output Data Flow :

Trip and Driver Information to Carpool Database

Type of Process :

☒ Online ☐ Batch ☐ Manual

Subprogram/Function Name:

Process Logic:

RETRIEVE Year of Study from User Database

IF Year of Study > 1

 THEN READ Ride Offer Details from Carpool Driver

 IF destination is not empty AND Available Seats greater than 0

 WRITE Trip and Driver Information to Carpool Database

 ELSE

 SEND error message to Carpool Driver

 ENDIF

ELSE

 SEND error message to Student

ENDIF

Refer to: Name: _____

☒ Structured English ☐ Decision Table ☐ Decision Tree

Unresolved Issues: -

Number : 8.2 Name : Handle Carpool Request Description : Shows available rides to the student, handles their join request, and sends the passenger's contact info and a match notification to the driver.	
Input Data Flow : Ride Offer List from Carpool Database Join-trip Request from Student	
Output Data Flow : Passenger Match Notification to Carpool Driver Passenger Contact Information to Carpool Driver Trip Booking & Acceptance Details to Process 8.3	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ Ride Offer List from Carpool Database READ Join-trip Request from Student IF Join-trip Request matched with Ride Offer List THEN SEND Passenger Match Notification to Carpool Driver SEND Passenger Contact Information to Carpool Driver CREATE Trip Booking & Acceptance Details SEND Trip Booking & Acceptance Details to Process 8.3 ELSE SEND "No matching rides found" to Student ENDIF	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 8.3 Name : Process Matching of Carpool Description : Processes the booking details, sends the driver's vehicle info to the student, and sends matched booking details to the next process.	
Input Data Flow : Trip Booking & Acceptance Details from Process 8.2	
Output Data Flow : Matched Driver Details to Student Matched Booking Details to Process 8.4	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ Trip Booking & Acceptance Details from Process 8.2 SET Matched Driver Details with Driver Contact Number, Pickup Time and Trip Fare SEND Matched Driver Details to Student SEND Matched Booking Details to Process 8.4	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

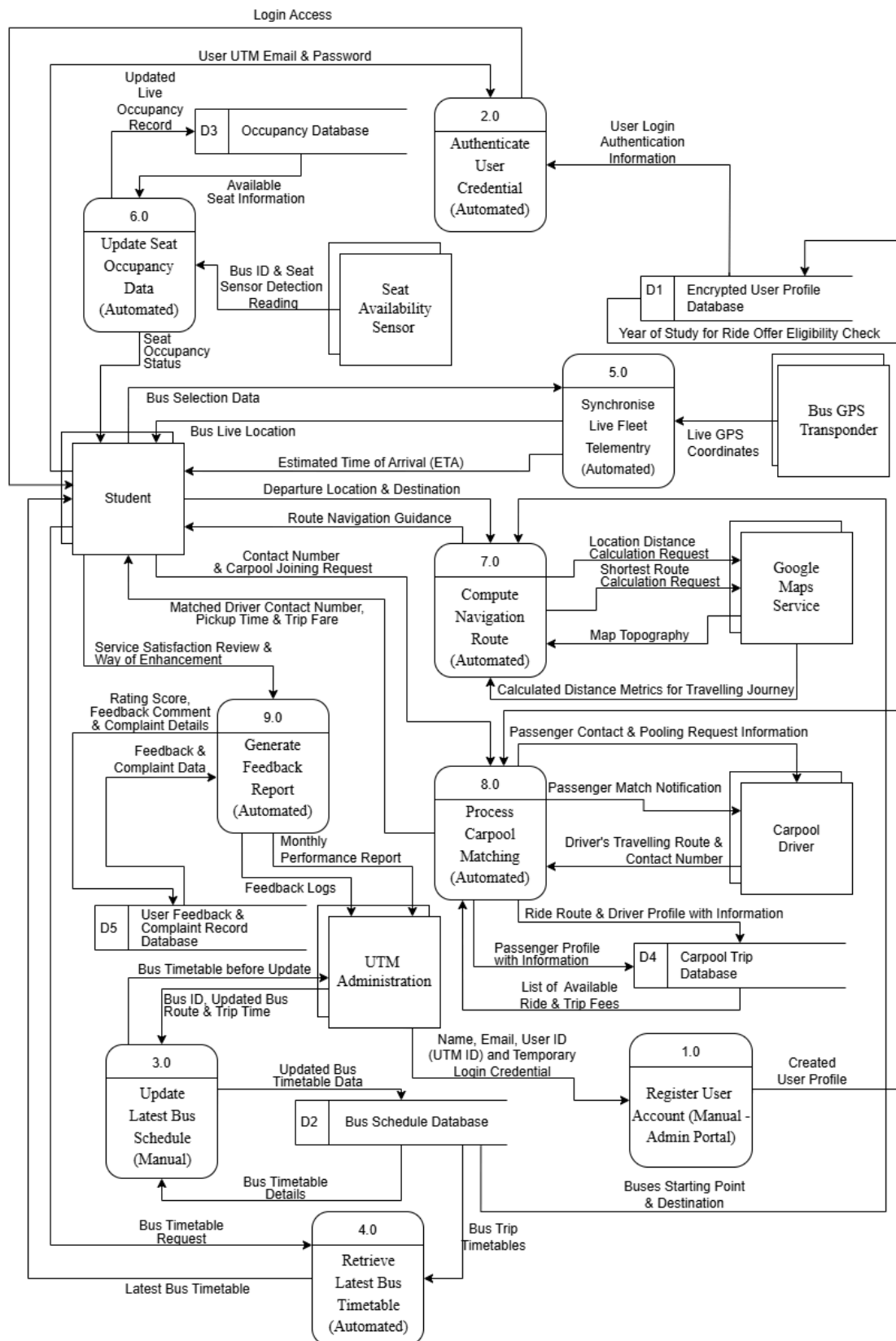
Number : 8.4 Name : Finalize Trip Records Description : Saves the final confirmed trip and passenger information into the carpool database for record keeping.	
Input Data Flow : Matched Booking Details from Process 8.3	
Output Data Flow : Passenger Information to Carpool Database	
Type of Process : ☒ Online ☐ Batch ☐ Manual	Subprogram/Function Name:
Process Logic: READ Matched Booking Details from Process 8.3 WRITE Passenger Information to Carpool Database	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

Number : 9.0 Name : Generate Feedback Report Description : Continuously receive and save live student trip feedback into the database while the system is operational, and handles requests to retrieve stored feedback data to generate performance reports and show feedback log to UTM Administration.	
Input Data Flow : Trip Feedback Data from Student Feedback & Complaint Data from Feedback Database	
Output Data Flow : Feedback Records to Feedback Database Feedback Logs to UTM Administration Monthly Performance Report to UTM Administration	
Type of Process : ☐ Online ☒ Batch ☐ Manual	Subprogram/Function Name:
<pre> DO WHILE system is operating READ Trip Feedback Data from Student IF Trip Feedback Data is received THEN CREATE New Feedback Record WRITE New Feedback Record to Feedback Database MOVE Feedback Data to Feedback Logs SEND Feedback Logs to UTM Administration ELSE continue ENDIF IF System Date is End of Month AND Current Time is EQUAL to "23:59" THEN READ Feedback & Complaint Data from Feedback Database COMPILE historical feedback trends and complaints from that month CREATE Monthly Performance Report SEND Monthly Performance Report to UTM Administration ELSE continue ENDIF ENDDO </pre>	
Refer to: Name: _____ ☒ Structured English ☐ Decision Table ☐ Decision Tree	
Unresolved Issues: -	

7.0 Physical System Design

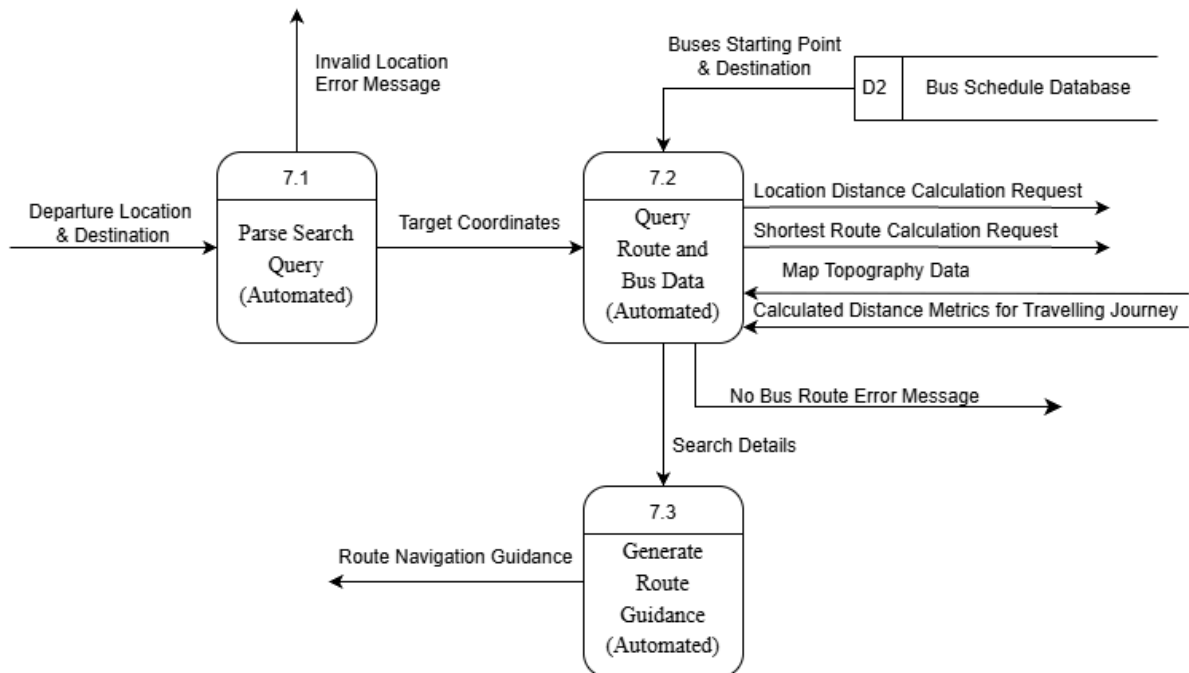
7.1 Physical DFD TO-BE System

Diagram 0

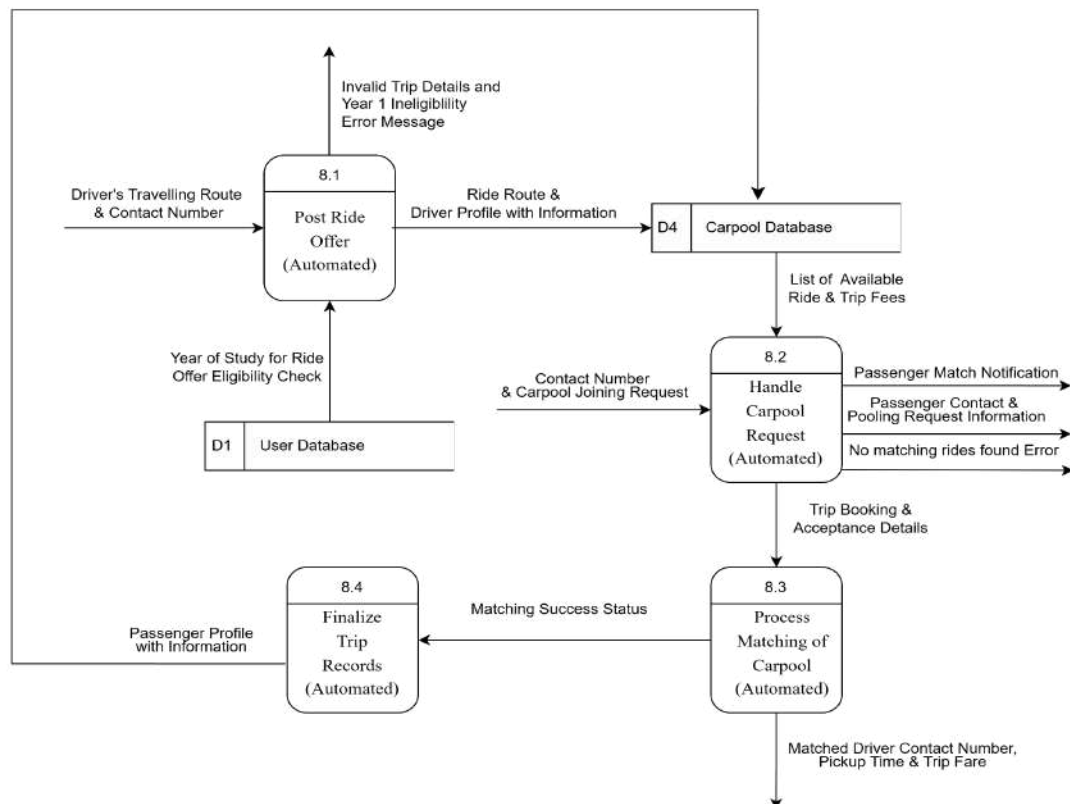


Child Diagram

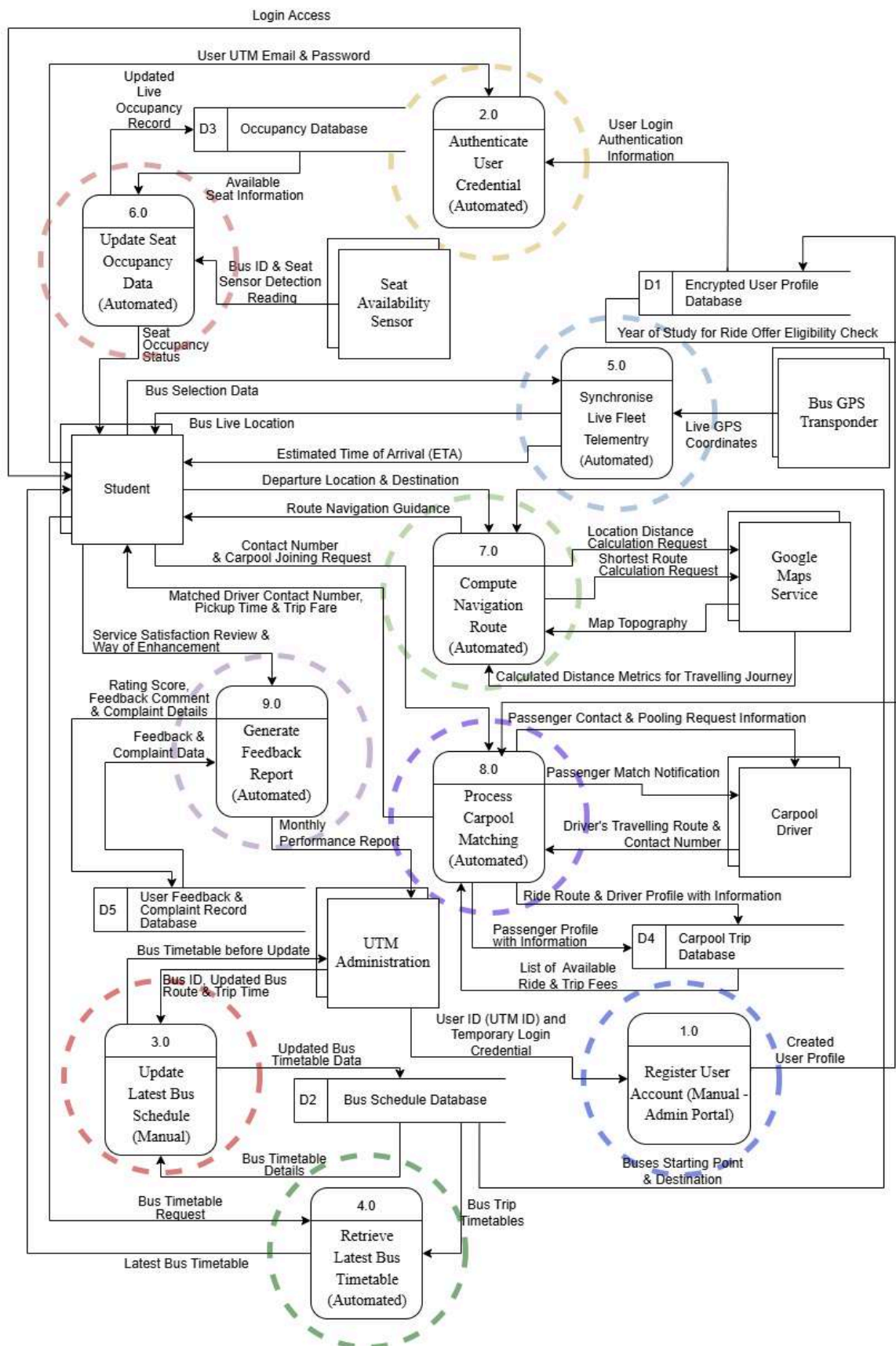
Process 7.1, 7.2, 7.3 - Route Guidance



Process 8.1, 8.2, 8.3, 8.4 - Carpool



Partitioning



CRUD Matrix

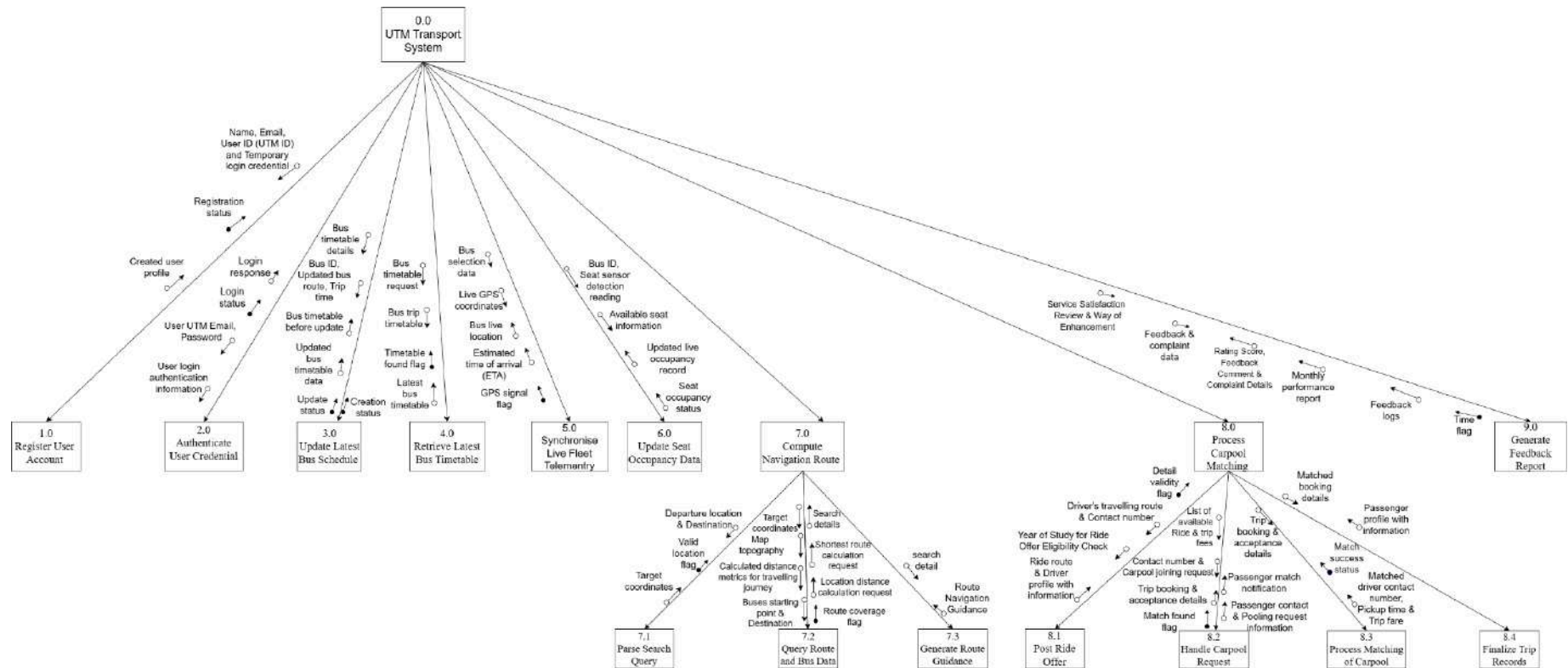
Data Store	User Database	Bus Schedule Database	Occupancy Database	Carpool Database	Feedback Database
Process					
Register User Account	CU				
Authenticate User	R				
Manage Bus Schedule		CUD			
Retrieve Bus Schedule		R			
Track Bus Live Location					
Handle Seat Occupancy			CRU		
Parse Search Query					
Query Route and Bus Data		R			
Generate Route Guidance					
Post Ride Offer				C	
Handle Carpool Request				CR	
Process Matching of Carpool					
Finalize Trip Records				U	
Generate Feedback Report					CR

Event Response Table

Event	Source	Trigger	Activity	Response	Destination
User registration initiated	UTM Administration	User registration details	Create new user profile Store new user profile	Account Creation Success Status	UTM Administration
User login	Student	Login details	Find user information. Verify credentials. Send login response	Login Success & Dashboard / Invalid Credentials Error	Student
UTM Admin add new bus schedule	UTM Administration	Bus schedule information	Store a new bus schedule.	Schedule Update Success Notification	UTM Administration
Student request bus schedule	Student	Bus timetable requirement	Retrieve the latest bus timetable.	Latest Bus Timetable	Student
Student select bus to get live location	Student	Bus selection	Get live GPS coordinates. Calculate ETA	Bus Live Location Estimated Time of Arrival (ETA)	Student
Sensor detects seat occupancy change	Seat Availability Sensor	Current Seat occupancy data	Receive new sensor reading Store occupancy record	Seat Occupancy Status.	Student
Student search location	Student	Location search	Parse target coordinates. Retrieve available bus routes and request	Location Distance Request. Route Calculation Request	Google Maps Service

			mapping data .		
Google Maps returns routing data	Google Maps Service	Map topography data Calculated distance metrics	Combine map metrics with bus route data. Generate and send route navigation guidance.	Route Navigation Guidance	Student
Driver offer ride	Carpool Driver	Ride offer details	Store ride offer details.	Ride Offer Posted Successfully confirmation	Carpool Driver
Student request join trip	Student	Join- trip request	Process the match request Send passenger information and match information to driver Send match information to student Store passenger information	Passenger contact information, passenger match notification Matched driver details	Carpool Driver Student
Student submit feedback	Student	Trip Feedback Data	Create and store new feedback record	Feedback Submission Success Message	Student
System generates monthly report		Temporal, monthly	Retrieve accumulated feedback records and generate the performance report.	Monthly Performance Report	UTM Administration

Structure Chart



System Architecture

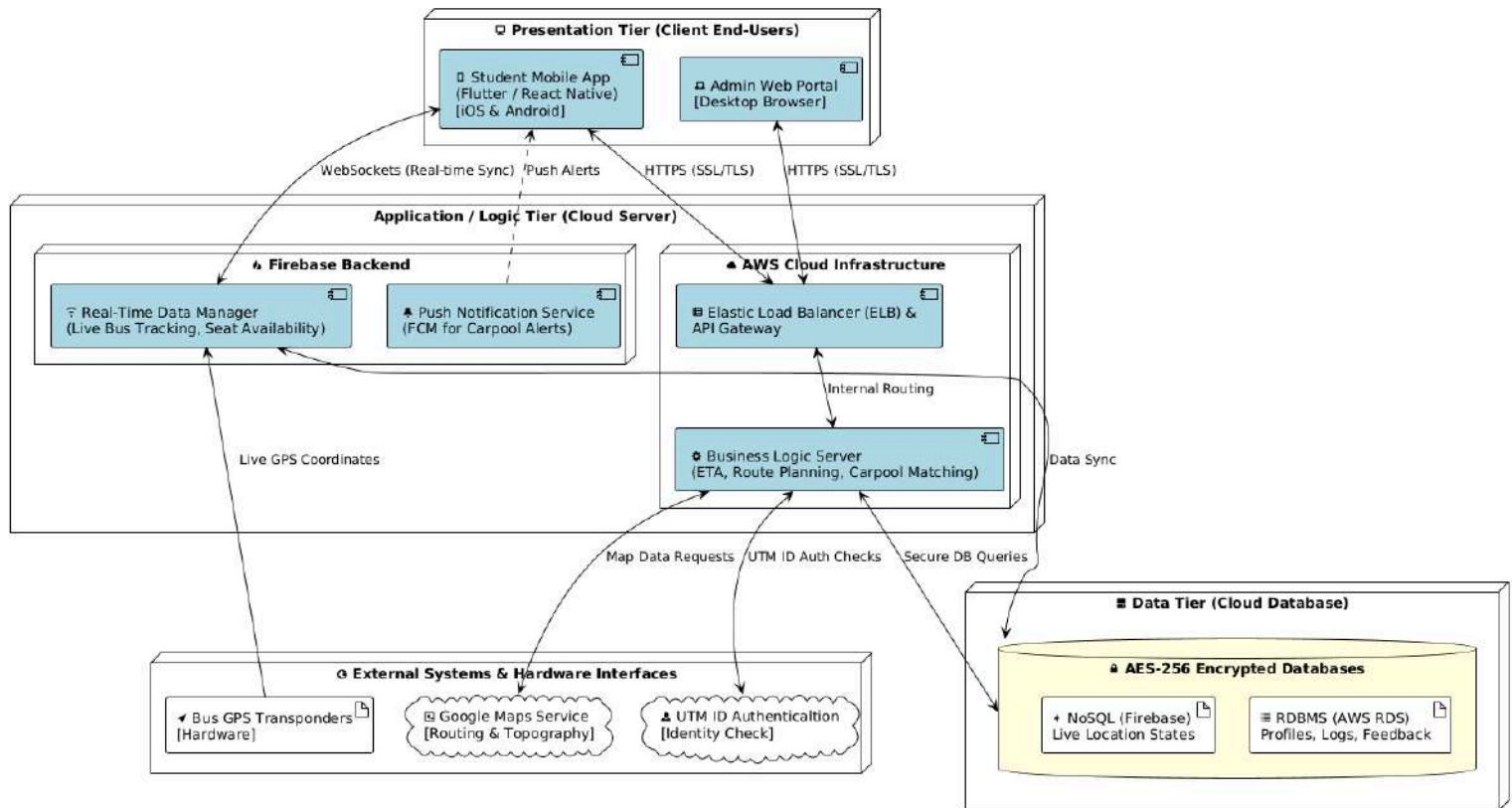


Figure: Overall System Architecture Diagram

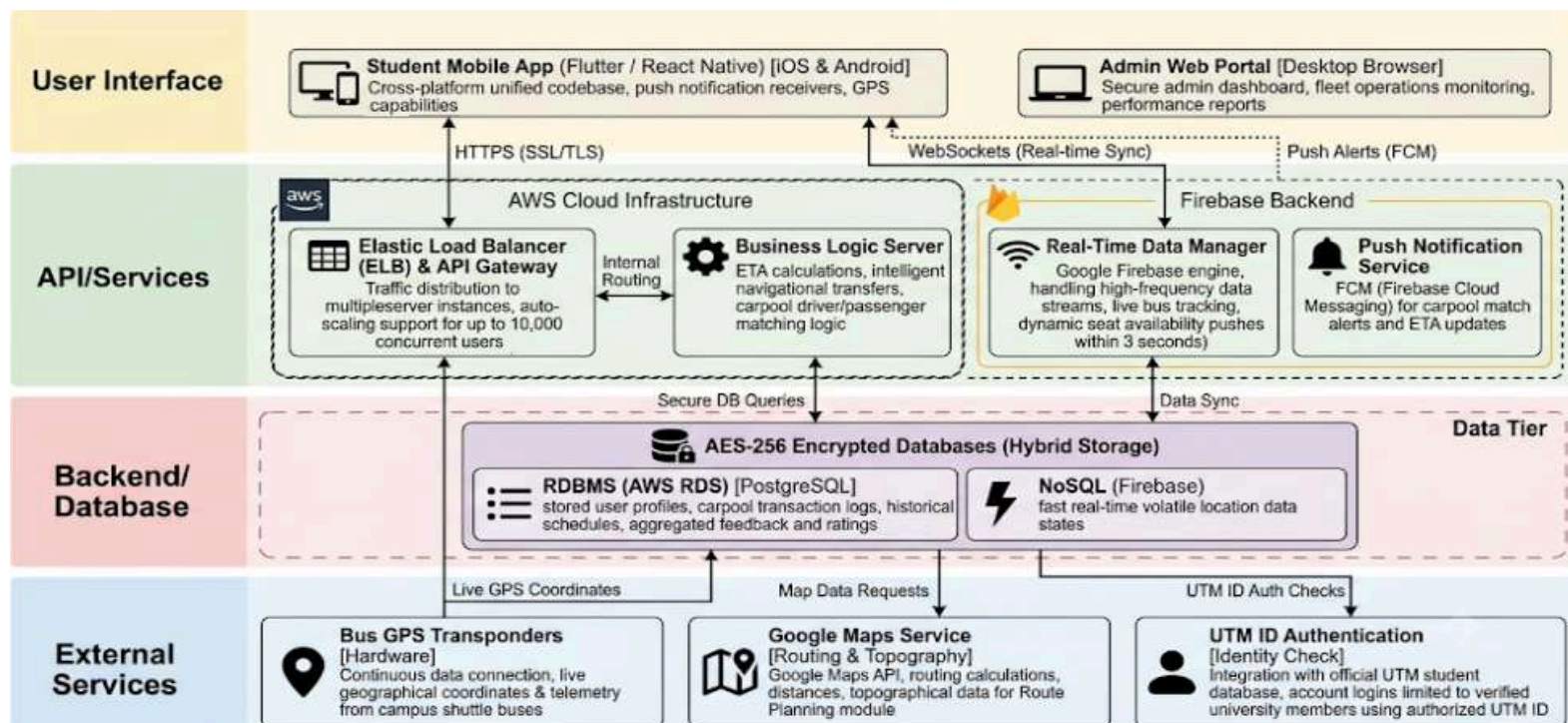


Figure: System Architecture Diagram According to Layer

I. Architecture Overview

UTM Transport App employs a robust, cloud-based, Three-Tier Client-Server Architecture. This structural design guarantees real-time data synchronization, dynamic scalability during peak campus transit hours, and high system availability. By decoupling the presentation, application, and data layers, the architecture ensures maintainability and allows for seamless integration with external hardware and third-party services. All communications between tiers are strictly secured by encryption to protect sensitive data in transit.

II. Presentation Tier (Client/User Interface)

This tier represents the physical end-user environment, ensuring cross-device compatibility and a frictionless user experience.

- **Student Mobile Application(Smartphones):** The primary interface for students utilizes cross-platform software frameworks such as Flutter or React Native. This allows a single, unified codebase to run natively on both iOS and Android operating systems, leveraging the device's built-in internet, push notification receivers, and GPS capabilities without requiring additional hardware purchases by the students.
- **Administrator Web Portal:** The UTM Transport Administrator and IT team interact with the system via a secure, web-based admin dashboard. This interface runs on standard desktop web browsers and allows administrators to monitor network health. Overseeing live fleet operations, and generating monthly performance reports.

III. Application Tier (Server Side)

This tier acts as the brain of the system, hosted entirely on cloud infrastructure to efficiently manage processing and live data distribution.

- **Cloud Hosting and Load Balancing:** Amazon Web Services (AWS) provides the scalable cloud computing infrastructure. AWS allows the system to automatically scale its processing resources. To seamlessly support a huge amount of concurrent users without performance degradation during peak hours, the system utilizes an API Gateway and an Elastic Load Balancer (ELB) to evenly distribute incoming traffic across multiple server instances.

- Real-Time Data Streams and Notifications: Google Firebase serves as the real-time backend engine. Firebase is crucial for handling continuous, high-frequency data streams, ensuring that live bus tracking and dynamic seat availability are pushed to student devices within 3 seconds of a change. Additionally, Firebase Cloud Messaging (FCM) is used to send instant push notifications to users when a carpool match is found or an ETA changes.
- Business Logic Processing: The server side executes all core computational tasks. This includes calculating precise Estimated Times of Arrival (ETA) based on bus speeds and traffic, generating intelligent navigational transfers, and programmatically matching carpool drivers with passengers based on departure times and destination proximity.

IV. Data Tier (Database)

This tier is responsible for secure, persistent storage of all system records, operating within cloud databases protected by strict data encryption standards to comply with external privacy provisions.

- Hybrid Data Storage Solution: The database securely stores and retrieves essential operational entities using a hybrid approach. Firebase's NoSQL database handles the volatile, fast-moving real-time location data, while a Relational Database Management System manages structured, historical data. This includes verified user profiles, a digital log of all carpool transactions, historical bus schedule data, and aggregate student feedback and rating reports. Automated backup protocols are in place to ensure that the probability of data corruption upon failure remains below 1%.

V. External Systems and Hardware Interfaces

To provide full functionality, UTM Transport App integrates with several critical third-party systems and external hardware.

- Hardware (Bus GPS Transponders): Physical GPS transponders are installed directly onto all active campus shuttle buses. These hardware devices establish a continuous data connection with the Application Tier, feeding live geographical coordinates and telemetry to accurately plot the fleet on the digital campus map.
- External API (Google Maps Service): The system reliably interfaces with the Google Maps API. This third-party integration is utilised by the Route

Planning module to fetch accurate campus topographies, calculate walking and transit distances, and generate step-by-step route calculations.

- **Security and Seamless Authentication (UTM ID Integration):**The system's authentication layer integrates directly with the student's unique UTM ID. This is because this ID is assigned to every student upon enrollment and serves as a universal key for all campus applications, it is already securely logged in the central university database. By linking out application to this existing infrastructure, we maintain a strictly verified, high-trust ecosystem restricted only to active students, while making the login process fast, unified, and effortless.

8.0 System Wireframe

Student Specification:

Input Design:

1. Students will be asked to enter UTM email and Password to login the UTM Transport App. The login page also provides a Remember me checkbox. After clicking this, students do not need to input the login credential again when they want to use this app.



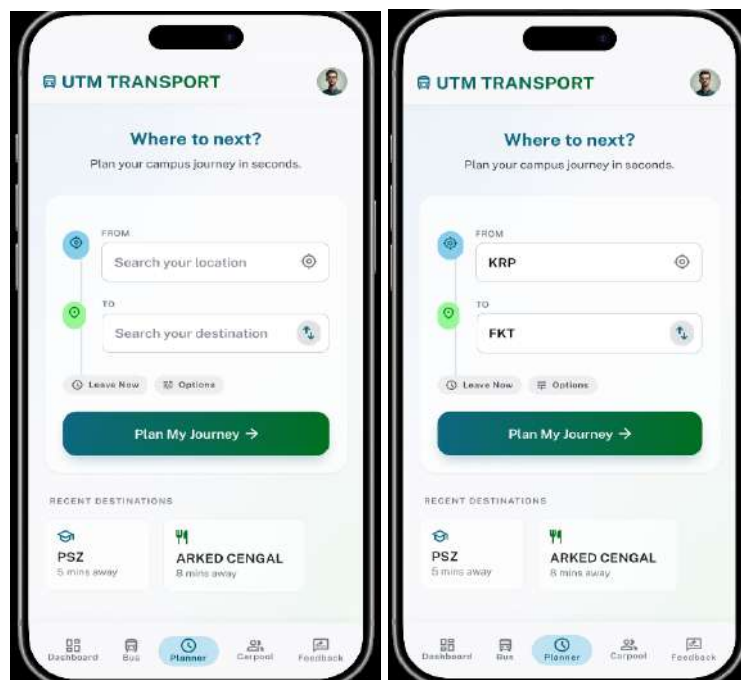
2. The dashboard provides two ways to access functions. The first one is the quick-access buttons for key features that show in the dashboard page and the other one is the navigation bar below to browse and go to corresponding pages.



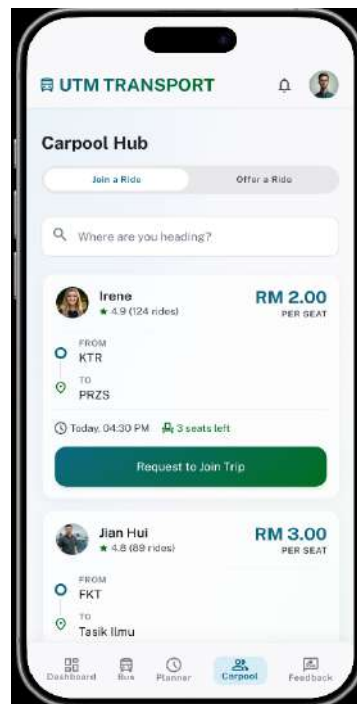
- Students can click the bus row for live tracking, or click the clock icon to view the timetable.



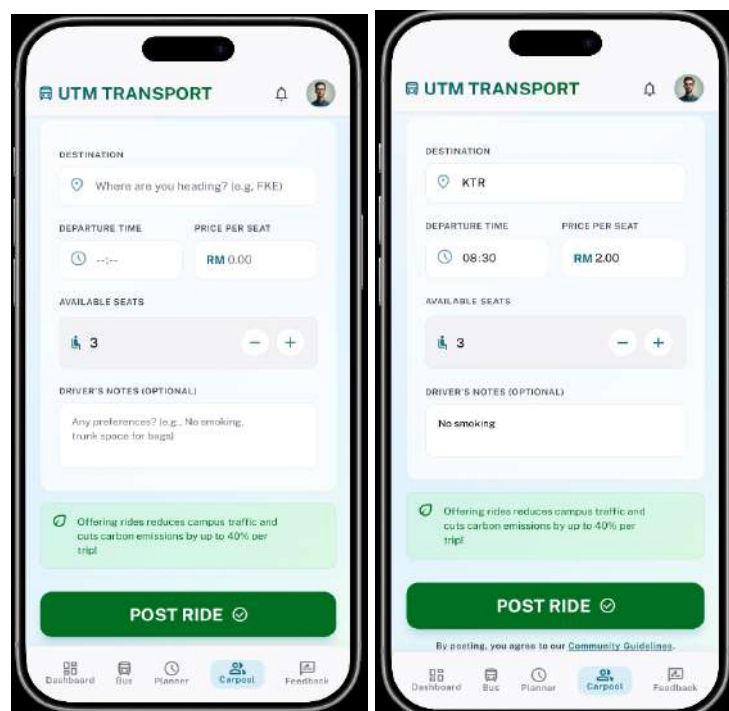
- Students can search the departure and destination here then click Plan My Journey for the Smart Route Planning feature.



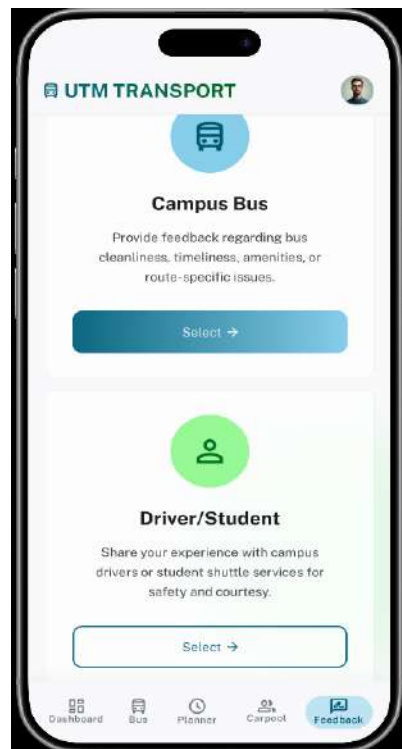
5. In the Carpool Hub, users can enter their destination in the search bar or click "Request to Join Trip" to join an existing ride. They can also click the notification bell icon at the top to view matched driver details. For drivers, they can click "Offer a Ride" to post a new ride.



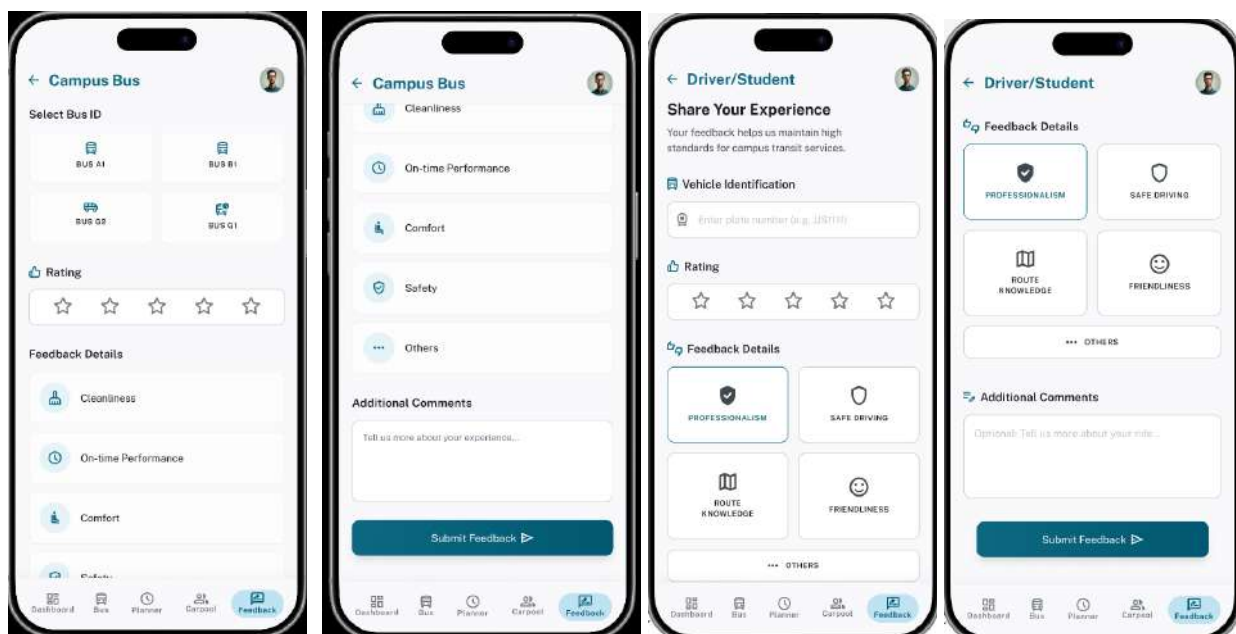
6. In the carpool hub, students as a driver can input the ride details and post ride, and click the notification bell icon to see the auto-matched passenger details.



7. Students can click the Select button to give the feedback.

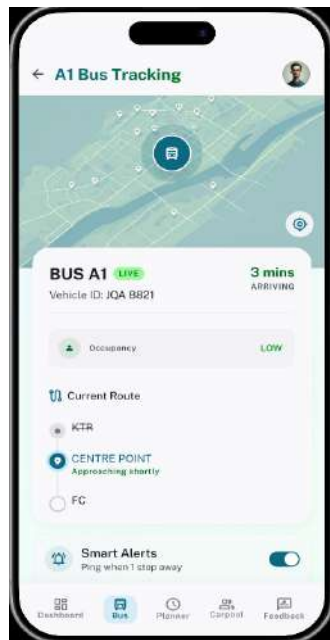


8. To give feedback to the campus buses, students can select the Bus ID, rate the bus driver, select feedback details and give additional comments, then click the Submit Feedback button. To give feedback to the driver, students can input the car plate number, rate the carpool driver, select feedback details and give comments, then submit feedback.

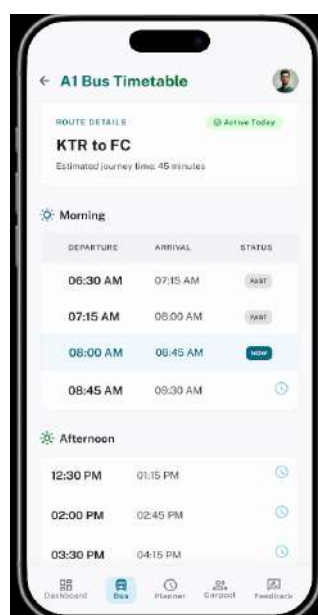


Output Design:

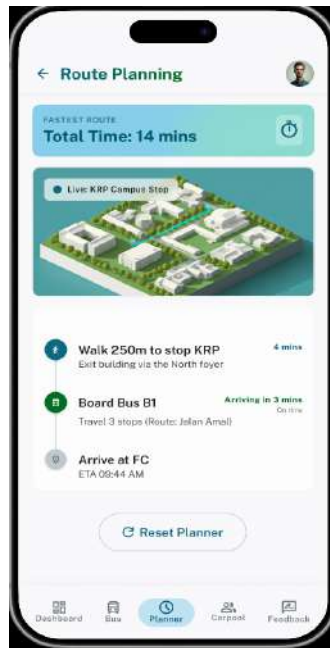
1. For live bus tracking, students can see the bus location and also the vehicle ID which is the plate number of the bus. They also can see the ETAs of the bus arriving at their location.



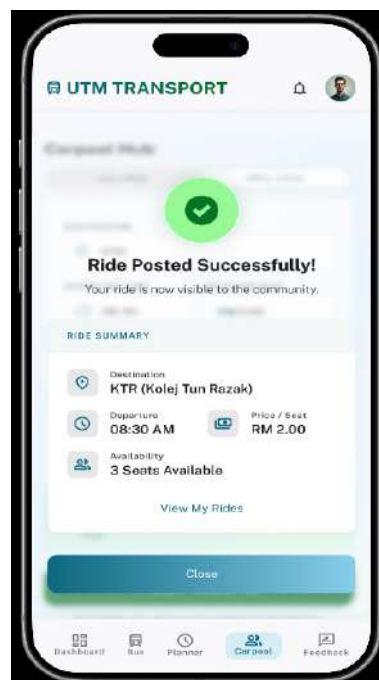
2. Students can see the timetable of the selected bus and it will show the route details. Students also can check whether the bus is active today or not and also can check the bus schedule is now on which period.



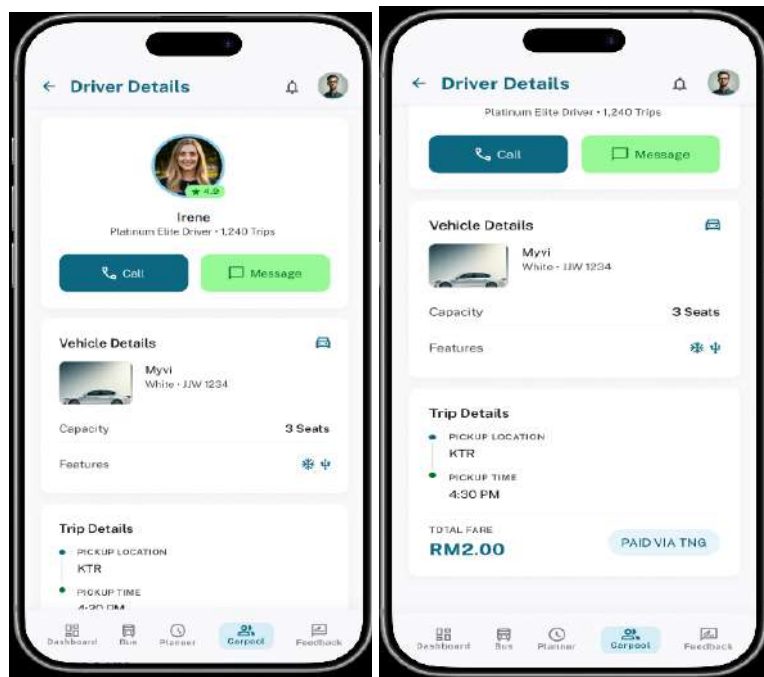
3. The interface displays the best route option and the total estimated travel time. Provides a map showing the student starting point and the physical path they need to take. It also shows the distance and time required to walk to the designated bus stop and show arriving time. The exact time the student will reach their final destination.



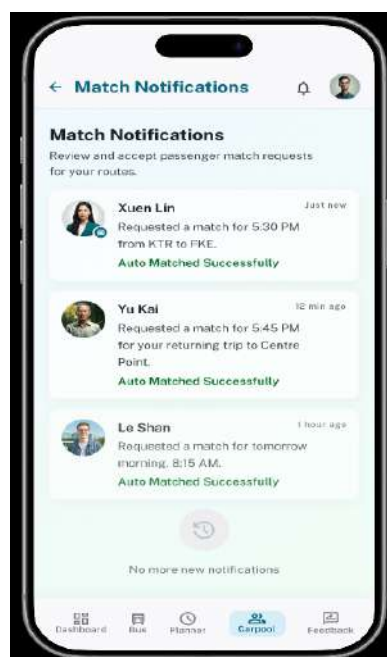
4. The screen displays a green tick and Ride Posted Successfully!. This message assures the student that the system has recorded their entry. It also shows the final drop-off point, the cost-sharing amount per passenger, total available spots and departure time.



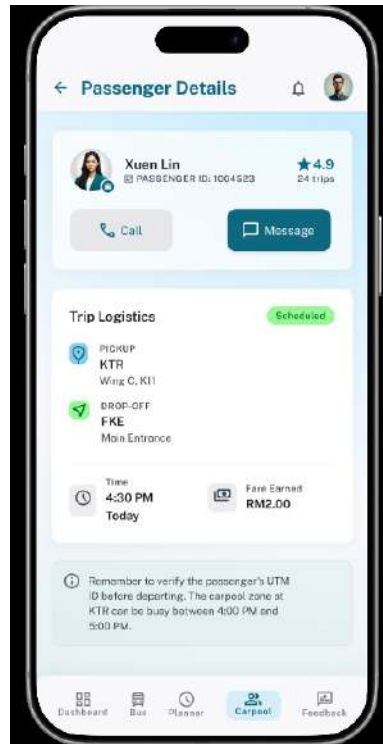
5. The screen displays the driver's photo, name, rating, and the trips that they have driven to ensure security and reliability before the ride. There are also vehicle details of the car, and also the trip details include the pickup location and pickup time.



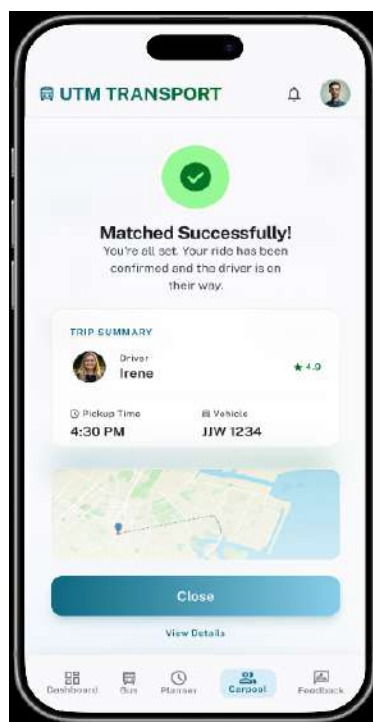
6. Displays a list of recent match activities sorted by time so drivers can quickly see the newest updates. It shows the student name and profile picture. It also shows the specific time and route.



7. It displays the passenger profile picture, name, rating, and the id of the passenger. It also displays the pickup and drop-off point of the passenger, the scheduled pickup time and the expected money earned from the passenger for completing the trip.



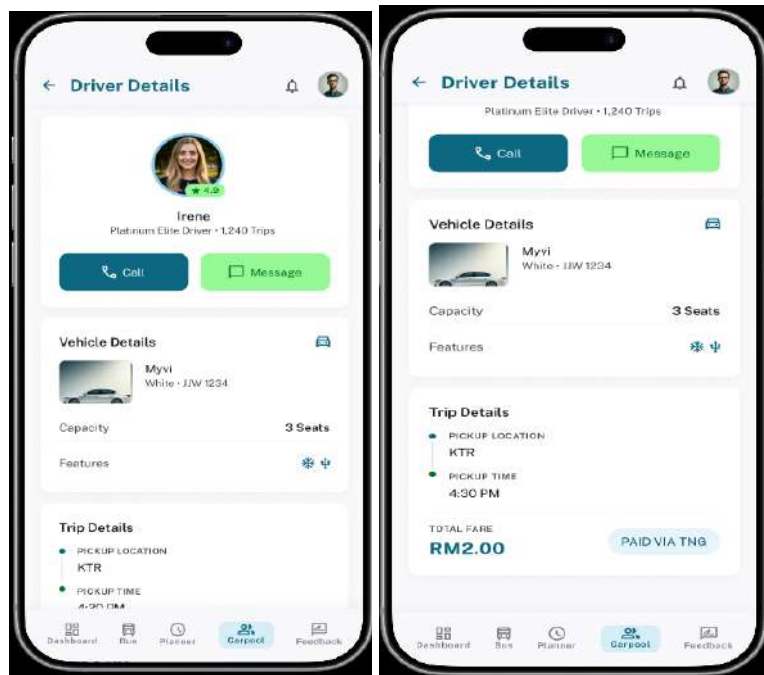
8. The screen displays the confirmation of the request to join the ride and shows a trip summary that includes the driver and ride information.



9. The screen displays the confirmation of the feedback and a unique reference ID is generated at the bottom. This provides the student with a formal record in case they need to follow up on a specific issue later.



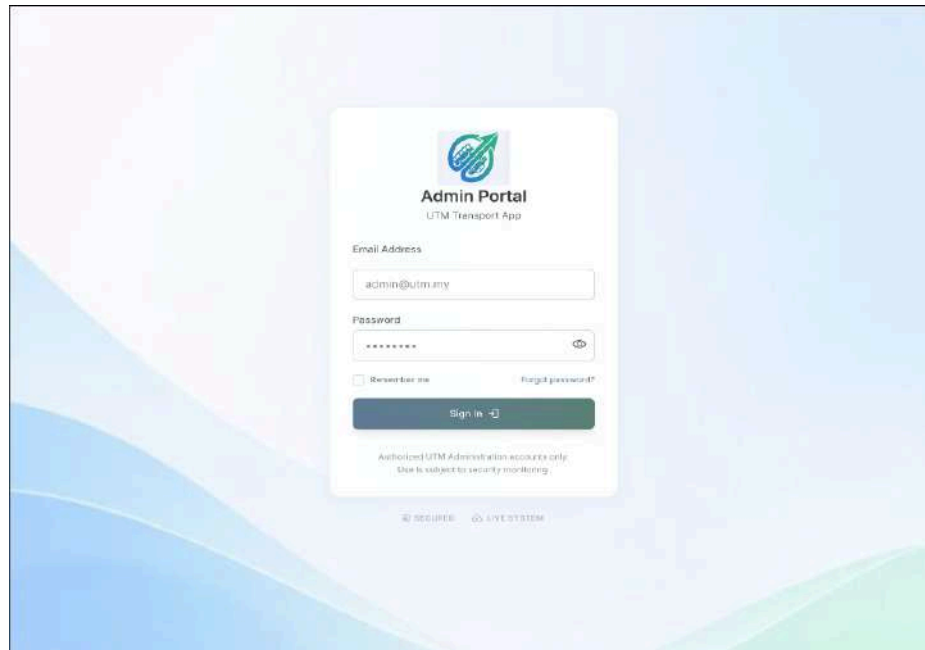
10. The student profile displays their profile picture, name, email and the details of it like total ride, feedback, faculty, year of study and also their student ID.



Admin Specification:

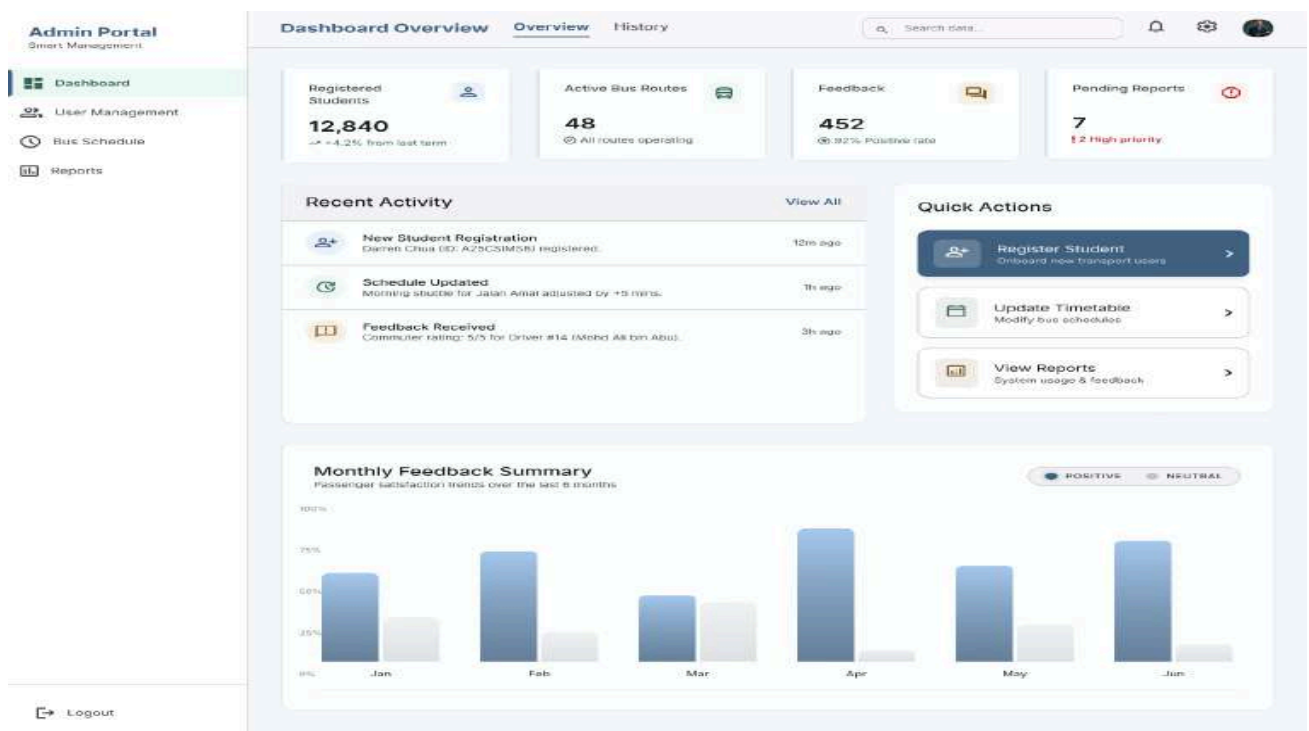
Input design:

1. Admin can enter an admin email and password to login the Admin Portal of the UTM Transport App. After entering all the login credentials, the admin can click the Sign In button to get into the Admin Portal.



The image shows a login form for the 'Admin Portal' of the 'UTM Transport App'. The form is centered on a background with abstract blue and green waves. At the top of the form is a logo consisting of a green leaf-like shape inside a blue circle, with the text 'Admin Portal' and 'UTM Transport App' below it. The form contains two input fields: 'Email Address' with the value 'admin@utm.my' and 'Password' with masked characters '*****'. Below the password field are two links: 'Remember me' (with an unchecked checkbox) and 'Forgot password?'. A dark green 'Sign in' button with a right-pointing arrow is positioned below these links. At the bottom of the form, there is a disclaimer: 'Authorized UTM Administration accounts only. Use is subject to security monitoring.' and two small icons labeled 'SECURE' and 'LIVE SYSTEM'.

2. The sidebar on the left side, there is user management, bus schedule and also Reports. Admin can click one of them and this will redirect them to the respective page that they have clicked. There is also a small logout function at the left bottom, if the admin wants to logout from this app then they can click it and it will redirect them to the login page. At the quick actions part, there are Register Students, Update Timetables and View Reports. This acts as a quick-access button that Admin can directly click the button and redirect to the page that performs this function.



3. To register a new account for students, admin can press the plus button and the right bottom. After that it will pop out the account registration panel, then the admin has to enter the full name, student ID, UTM official email into the specific blank that is available. After that, the admin can generate the temporary password for the student and also click the auto-activate account for automatic activation. After ensuring all the information is entered correctly, then the admin can click the Create Account button then the student account will be registered.

The screenshot displays the 'Admin Portal' interface for 'User Management'. The left sidebar contains navigation links: Dashboard, User Management (selected), Bus Schedule, and Reports. The main content area is titled 'User Management' with tabs for 'Overview' and 'History'. A search bar 'Search students...' is located at the top right. Below the search bar is a 'Registered Students' table with columns: No., Name, Student ID, Email, and Joined Date. The table lists three students: Ethan Johnson, Sarah Miller, and David Chen. To the right of the table is a 'New Account' panel with fields for Full Name, Student ID, University Email, and Temporary Password, along with an 'Auto-activate account' toggle and a 'Create Account' button. A note at the bottom of the panel states: 'Creating an account will trigger an automated enrollment email to the student with their temporary login credentials and the Campus Transport mobile app guide.' A '+ button is located at the bottom right of the interface.

No.	Name	Student ID	Email	Joined Date
01	EJ Ethan Johnson	A25CS6666	ethen@graduate.utm.my	Oct 12, 2025
02	SM Sarah Miller	A25CS4890	sarah@graduate.utm.my	Nov 05, 2025
03	DC David Chen	A25CS1234	david@graduate.utm.my	Jan 18, 2026

Showing 3 of 1,240 students

New Account
Register a student manually

Full Name
John Doe

Student ID
A25CS0001

University Email
john@graduate.utm.my

Temporary Password
TP_2047_#X

Auto-activate account ☒

Create Account

Creating an account will trigger an automated enrollment email to the student with their temporary login credentials and the Campus Transport mobile app guide.

4. The purpose of this bus schedule management interface is to allow the admin to manage and monitor the current bus schedule. Admin can manage the bus timings and the route efficiently.

The screenshot displays the 'Admin Portal' for 'Smart Management'. The left sidebar contains navigation links: 'Dashboard', 'User Management', 'Bus Schedule' (highlighted), and 'Reports'. At the bottom of the sidebar is a 'Logout' button. The main content area is titled 'Bus Schedule Management' and includes tabs for 'Overview' and 'History'. A search bar labeled 'Search schedules...' is present. Below the search bar are filters for 'Route: All Routes', 'Day: Everyday', and 'Date: mm/dd/yyyy', along with a '+ Add New Schedule' button. The central part of the interface is a table with the following columns: 'BUS ID', 'ROUTE', 'DEP / ARR', 'TIMES', 'DAYS', 'STATUS', and 'ACTIONS'. The table lists four bus schedules, all with a status of 'ACTIVE'. The first schedule, #A1, is for 'Jalan Amal' (KTR to FC) with times 08:00-08:45 and days M, T, W, T, F. The second, #B1, is for 'Jalan Cengal' (KTHD to FC) with times 09:00-09:30 and days M, W, F. The third, #C1, is for 'Tasik Ilmu' (KRP to FKT) with times 10:15-10:50 and days S, S. The fourth, #D1, is for 'Lingkeran Ilmu' (KRP to FKE) with times 07:30-08:10 and days M, T, W, T, F. Each row has edit and delete icons in the 'ACTIONS' column. At the bottom of the table, it says 'Showing 1 to 4 of 24 schedules' with pagination controls for pages 1, 2, and 3.

BUS ID	ROUTE	DEP / ARR	TIMES	DAYS	STATUS	ACTIONS
#A1	Jalan Amal	KTR → FC	08:00 - 08:45	M, T, W, T, F	ACTIVE	
#B1	Jalan Cengal	KTHD → FC	09:00 - 09:30	M, W, F	ACTIVE	
#C1	Tasik Ilmu	KRP → FKT	10:15 - 10:50	S, S	ACTIVE	
#D1	Lingkeran Ilmu	KRP → FKE	07:30 - 08:10	M, T, W, T, F	ACTIVE	

5. For adding a new bus schedule, the admin has to click the add new schedule button at the right top of the page. After clicking it, then it will pop out the panel. Then the admin has to enter Bus ID, Route name, Departure and Destination point and the departure and arrival time. Admin should choose the days that the bus is active by clicking the day button. Admin should also click the active status to make sure that the bus schedule is activating after being added. After all the information is entered, the admin can click the Save Schedule button for saving and this will be done for the adding schedule process.

Admin Portal
Smart Management

- Dashboard
- User Management
- Bus Schedule**
- Reports

Bus Schedule Management | Overview | History | Search

Route: All Routes | Day: Everyday | Date: mm/dd/yy

BUS ID	ROUTE	DEP / ARR	TIMES
#A1	Jalan Amal	KTR → FC	08:00 - 08:45
#B1	Jalan Cengal	KTHO → FC	09:00 - 09:30
#C1	Tasik Ilmu	KRP → FKT	10:15 - 10:50
#D1	Lingkar Ilmu	KRP → FKE	07:30 - 08:10

Showing 1 to 4 of 24 schedules

Add New Schedule
Create a new operational bus route entry.

Bus ID: #E1

Route Name: Kolej Tun Razak

Departure Point: Centre Point | Destination: KTR

Departure Time: 08:30 | Arrival Time: 09:15

Days Active: Mon, Tue, Wed, Thu, Fri, Sat, Sun

Active Status: Enable this schedule immediately ☒

Save Schedule | Cancel

Logout

6. To edit the bus schedule, the admin can click the pencil icon on the actions column. After clicking it, it will pop out the panel as shown below. Admin can change the Bus ID, Route name, Departure and Destination point, the departure and arrival time, the status of the bus and the days that the bus is active. After making sure all the changes have entered, then the admin can click the Save Changes button, then the bus schedule will be changed and updated to the latest version.

Admin Portal
Smart Management

- Dashboard
- User Management
- Bus Schedule**
- Reports

Bus Schedule Management | Overview | History | Search schedule

Route: All Routes | Day: Everyday | Date: mm/dd/yyyy

BUS ID	ROUTE	DEP / ARR	TIMES
#A1	Jalan Amal	KTR → FC	08:00 - 08:45
#B1	Jalan Cengal	KTHQ → FC	09:00 - 09:30
#C1	Tasik Ilmu	KRP → FKT	10:15 - 10:50
#D1	Lingkar Ilmu	KRP → FKE	07:30 - 08:10

Showing 1 to 4 of 24 schedules

Edit Schedule #A1 Jalan Amal

Bus ID: #A1 | Status: ☒ Active Route

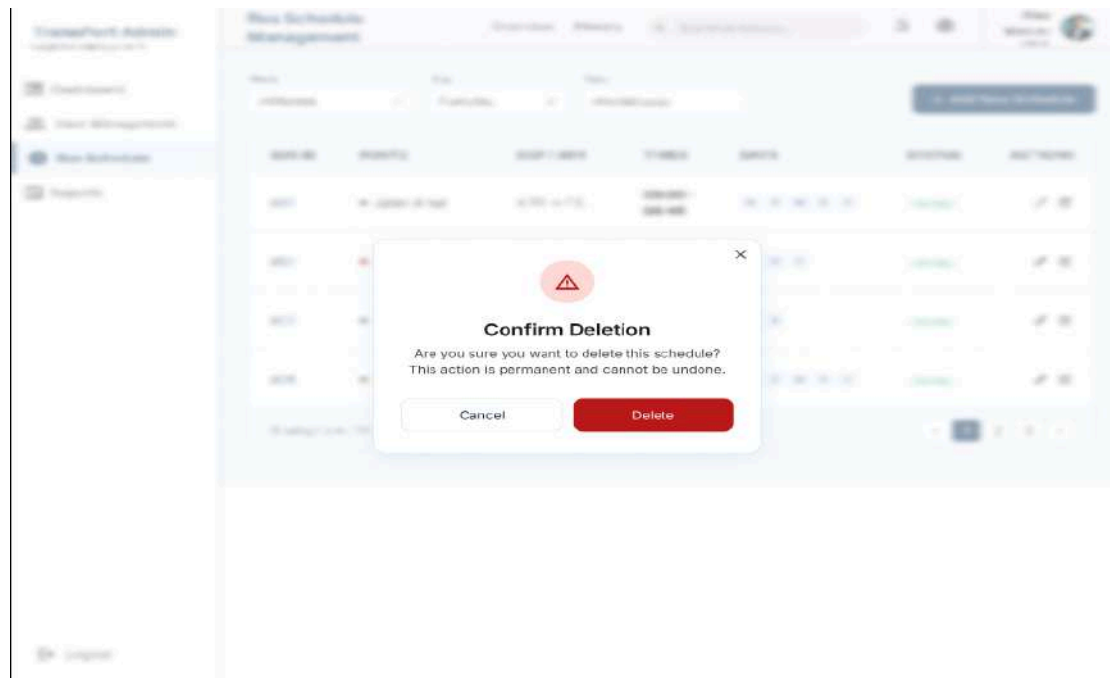
Route Name: Jalan Amal

Departure Point: KTR | Destination: FC

Departure Time: 08:10 AM | Arrival Time: 08:30 AM

Days Active: ☒ Mon ☐ Tue ☒ Wed ☐ Thu ☐ Fri ☐ Sat ☐ Sun

7. To delete the bus schedule, the admin should click on the rubbish bin icon beside the pencil icon. After clicking, the app will pop out an interface and it will display the message about asking for confirmation to delete the bus schedule. If the admin wants to delete, then click the Delete button, if they want to cancel the deletion, then just click the Cancel button.



Output Design:

1. The purpose of this interface is to show after the submission of a new user registration, the interface immediately updates the registered student list and also pop up the confirmation message about the student being successfully registered and added into the list.

The screenshot displays the 'Admin Portal' interface with a sidebar menu on the left containing 'Dashboard', 'User Management' (selected), 'Bus Schedule', and 'Reports'. The main content area is titled 'User Management' and includes tabs for 'Overview' and 'History'. A green notification banner at the top states 'New student added successfully'. Below this, a table titled 'Registered Students' lists four students with columns for No., Name, Student ID, Email, Joined Date, Status, and Actions. The status for all students is 'ACTIVE'. At the bottom, it indicates 'Showing 4 of 12,841 students' and features a '+ Add' button.

No.	Name	Student ID	Email	Joined Date	Status	Actions
01	EJ Ethan Johnson	A25CS888	ethan@graduate.utm.my	Oct 12, 2025	ACTIVE	
02	SM Sarah Miller	A25CS4896	sarah@graduate.utm.my	Nov 05, 2025	ACTIVE	
03	DC David Chen	A25CS1234	david@graduate.utm.my	Jan 18, 2026	ACTIVE	
04	JD John Doe	A25CS0001	john@graduate.utm.my	June 18, 2026	ACTIVE	

- The interface will pop up a green alert banner from the top center that notified the admin that the new bus schedule is successful. The new bus schedule will appear straight after the admin saves it and admin can get the latest version of the bus schedule.

Admin Portal
Smart Management

Dashboard

User Management

Bus Schedule

Reports

Logout

Bus Schedule Management

OverviewHistory

Search schedules...

🔔⚙️👤

New bus schedule added successfully

Route: All Routes

Day: Everyday

Date: mm/dd/yyyy

+ Add New Schedule

BUS ID	ROUTE	DEP / ARR	TIMES	DAYS ACTIVE	STATUS	ACTIONS
#A1	Jalan Amal	KTR → FC	08:00 - 08:45	M T W	ACTIVE	 
#B1	Jalan Cengal	KTHO → FC	09:00 - 09:30	M T	ACTIVE	 
#C1	Tasik Ilmu	KRP → FKT	10:15 - 10:50	W T	ACTIVE	 
#D1	Lingkarani Ilmu	KRP → FKE	07:30 - 08:10	F	ACTIVE	 
#E1	Kolej Tun Razak	Centre Point → KTR	08:30 - 09:15	M T W T F	ACTIVE	 

Showing 1 to 5 of 25 schedules

< 1 2 3 >

3. The interface will pop up a green alert banner from the top center that notified the admin that the bus schedule is successfully changed. As we can see, the times of the Bus A1 have changed from the previous one.

Admin Portal
Smart Management

Dashboard

User Management

Bus Schedule

Reports

Bus Schedule Management

OverviewHistory

Search schedules...

Bus schedule successfully changed

Route: All RoutesDay: EverydayDate: mm/dd/yyyyAdd New Schedule

BUS ID	ROUTE	DEP / ARR	TIMES	DAYS	STATUS	ACTIONS
#A1	Jalan Amal	KTR → FC	08:10 - 08:30	M T W T F	ACTIVE	 
#B1	Jalan Cengal	KTHO → FC	09:00 - 09:30	M W F	ACTIVE	 
#C1	Tesik Ilmu	KRP → FKT	10:15 - 10:50	S S	ACTIVE	 
#D1	Lingkarani Ilmu	KRP → FKE	07:30 - 08:10	M T W T F	ACTIVE	 

Showing 1 to 4 of 24 schedules

<123>

Logout

4. The interface will pop up a green alert banner from the top center that notified the admin that the bus schedule is successfully deleted. As we can see, the bus schedule of the Bus A1 has disappeared from the list.

TransPort Admin
Logistics Management

Dashboard
User Management
Bus Schedule
Reports

Bus Schedule Management Overview History

Route: All Routes Day: Everyday Date: mm/dd/yyyy + Add New Schedule

Schedule #A1 deleted successfully

BUS ID	ROUTE	DEP / ARR	TIMES	DAYS	STATUS	ACTIONS
#B1	Jalan Cengal	KTHO → FC	09:00 - 09:30	M W F	ACTIVE	
#C1	Tasik Ilmu	KRP → FKT	10:15 - 10:50	S S	ACTIVE	
#D1	Lingkar Ilmu	KRP → FKE	07:30 - 08:10	M T W T F	ACTIVE	

Showing 1 to 3 of 23 schedules

1 2 3

Logout

5. Admin can look at the feedback from the student from the feedback log. There is a category, rating, date of feedback and the status of it. Admin can get the reports by clicking the monthly performance reports bar beside the feedback log and it will redirect to the specific page.

Admin Portal
Smart Management

Dashboard

User Management

Bus Schedule

Reports

Feedback & Reports

OverviewHistory

Search data...

TOTAL FEEDBACK
1,284
+12% from last month

AVG RATING
4.8 ★★★★★
Based on 850 reviews

OPEN COMPLAINTS
14
Resolve pending >

Feedback Log

Monthly Performance Reports

USER	CATEGORY	RATING	DATE	STATUS
Alex Johnson	Bus Timeliness	★★★★★	June 24, 2026	Resolved
Sarah Miller	Driver Conduct	★★★★☆	June 23, 2026	Pending
99 "The driver on Jalan Amel was very impatient today. He didn't wait for me to sit down before accelerating, which felt dangerous for students. Otherwise, the bus was clean." AcknowledgeEscalate				
Marcus Chen	Driver Conduct	★★★★☆	June 22, 2026	Resolved

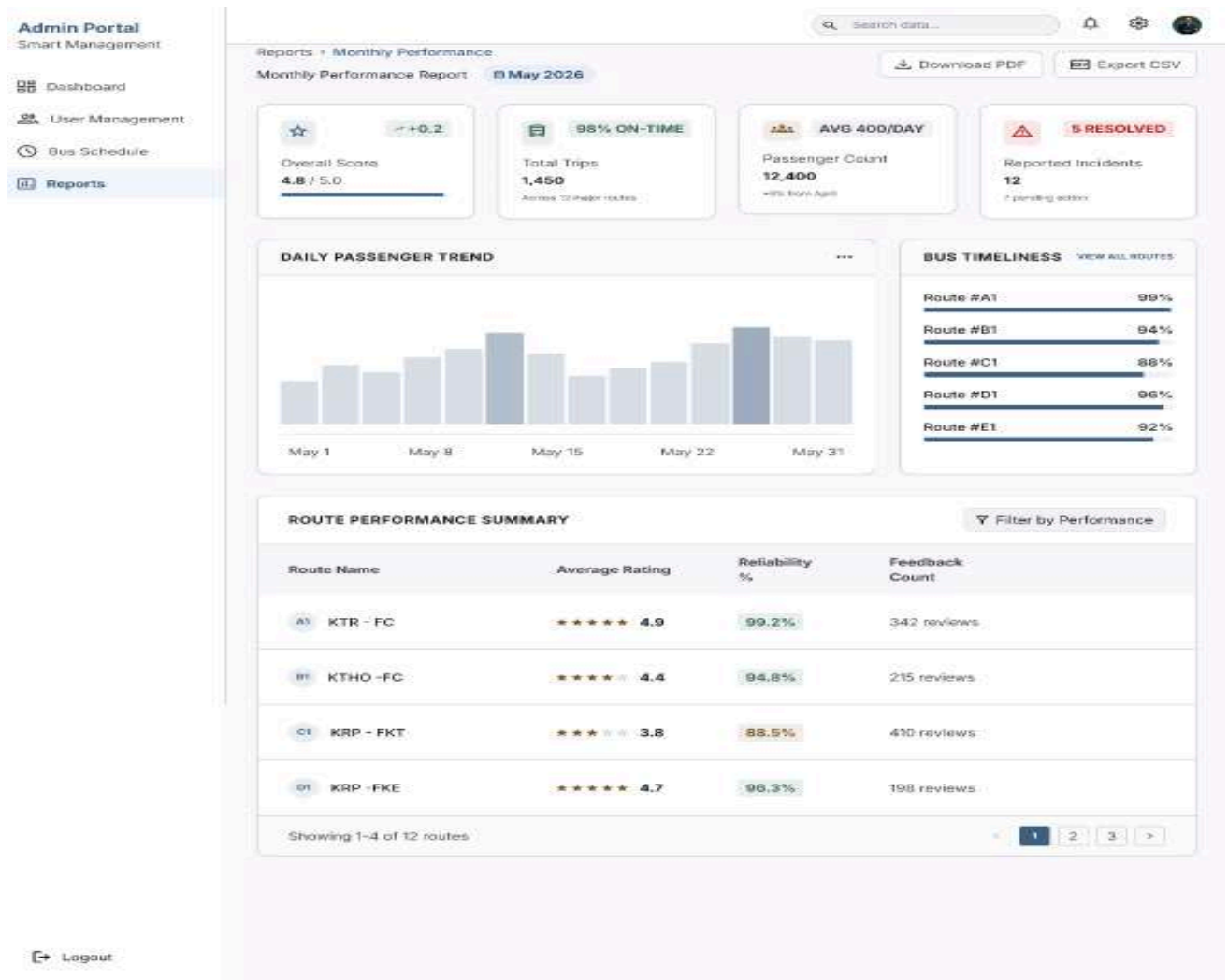
Showing 1-10 of 1,284 entries

<123>

Auto-Generation Notice: Monthly performance reports are automatically generated on the end of every month at 23:59 p.m. Previous months are archived and available for download.

Logout

6. After clicking the monthly performance reports bar, it will show this interface. This page is to let the admin look out each performance of the feature or function of the UTM transport app. Admin gets to know which bus is mostly used by students and what are the issues reported and how many of them are solved.



9.0 Summary of the Proposed System

The UTM Transport System is designed to resolve five campus transit pain points, they are bus arrival uncertainty, peak-hour seat shortages, unsafe carpooling, lack of route planning tools, and the absence of a proper feedback system. This project introduces a system to solve these problems. It includes secure user registration, live bus tracking with estimated time of arrival (ETA), real-time seat monitoring, official carpooling feature, smart route navigation and digital feedback system.

The system design ensures strict data boundaries and smooth information flow. User verification takes place through the internal registry. Real-time vehicle location updates continuously from GPS transponders. Passenger occupancy levels are tracked automatically via seat sensors. Moreover, the system provides matching rides for students and drivers. This system also integrated with google maps service to calculate the shortest route. All main storage units such as Carpool trip Database, Bus schedule Database and so on are designed consistently according to the overall architecture.

In conclusion, this system successfully solves the challenges posed by everyday delays in transit and administrative blind spots. By standardizing core processes, the system efficiently translates raw information from users and vehicles into accurate tracking and monthly performance checks. Ultimately, it is hoped that this application will not only minimize transit anxiety but also foster a more efficient campus community for all UTM students.